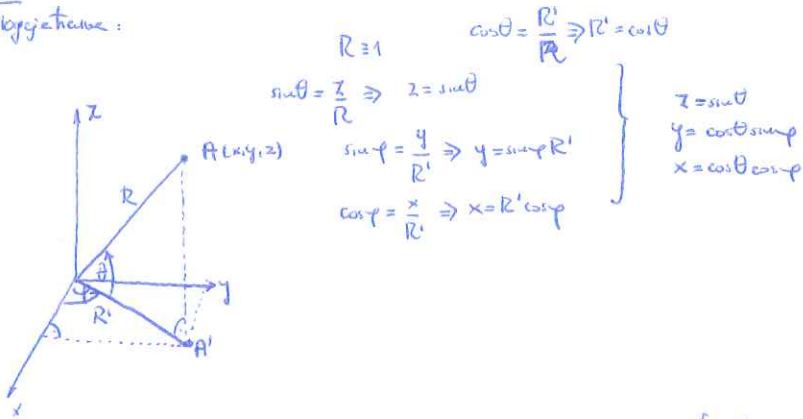


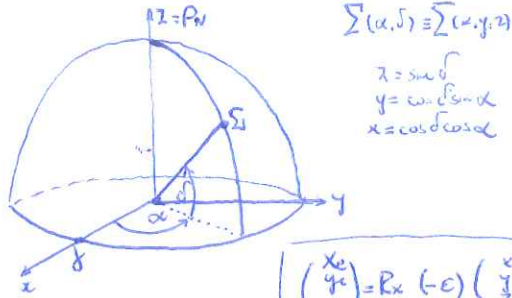
Општа астрономија 2, школска 2025/2026

Испод можете пронаћи сакупљене старе биљешке које се тичу опште астрономије 2 (у питању су задаци). Исте могу да садрже рачунске или друге грешке. Материјал треба да послужи као помоћ колегама који нису редовно похађали вјежбе.

Ротация:

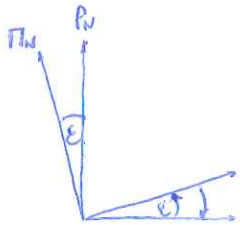


а) Преобразование евклидовых координат



$$\begin{pmatrix} x_e \\ y_e \\ z_e \end{pmatrix} = R_x(-\epsilon) \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = R_x(+\epsilon) \begin{pmatrix} x_e \\ y_e \\ z_e \end{pmatrix}$$



$$\begin{pmatrix} x_e \\ y_e \\ z_e \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \epsilon & \sin \epsilon \\ 0 & -\sin \epsilon & \cos \epsilon \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \text{— евклидовы в сферические}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \epsilon & -\sin \epsilon \\ 0 & \sin \epsilon & \cos \epsilon \end{pmatrix} \begin{pmatrix} x_e \\ y_e \\ z_e \end{pmatrix} \quad \text{— сферические в евклидовых}$$

$$x_e = x$$

$$y_e = \cos \epsilon y + \sin \epsilon z$$

$$z_e = -\sin \epsilon y + \cos \epsilon z$$

$$x = x_e$$

$$y = \cos \epsilon y_e - \sin \epsilon z_e$$

$$z = \sin \epsilon y_e + z_e \cos \epsilon$$

$$x_e = \cos \theta \cos \varphi$$

$$y_e = \cos \theta \cos \varphi \sin \lambda + \sin \theta \sin \lambda$$

$$z_e = -\sin \theta \cos \varphi \sin \lambda + \cos \theta \sin \lambda$$

$$x = \cos \theta \cos \varphi$$

$$y = \cos \theta \cos \varphi \sin \lambda - \sin \theta \sin \lambda$$

$$z = \sin \theta \cos \varphi \sin \lambda + \sin \theta \cos \lambda$$

Ано у гаче сферические евклидовые координаты, мате λ и β

$$\epsilon = 23^\circ 26' 24''.96$$

$$x = 0.9798520, y = -0.1741053, z = -0.0782447$$

иначе:

$$x_e = x = 0.9798520$$

$$y_e = \cos \epsilon y + \sin \epsilon z = -0.9798520$$

$$z_e = -y \sin \epsilon + z \cos \epsilon = 0.0002227$$

$$z_e = \sin \beta$$

$$x_e = \cos \beta \cos \lambda$$

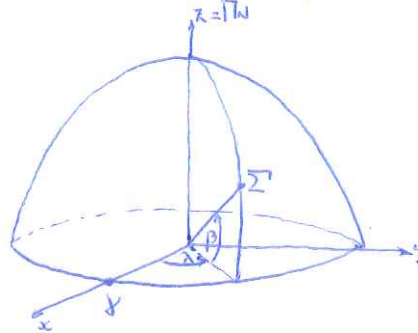
$$y_e = \cos \beta \sin \lambda$$

$$\frac{y_e}{x_e} = \tan \lambda = -0.19350313 \Rightarrow \lambda = 349^\circ 2' 41''.25$$

$$\frac{z_e}{\sqrt{x_e^2 + y_e^2}} = \frac{\sin \beta}{\sqrt{\cos^2 \beta \cos^2 \lambda + \cos^2 \beta \sin^2 \lambda}} = \frac{\sin \beta}{\cos \beta} = \tan \beta = 0.0002227$$

$$\Rightarrow \beta = 46''.625$$

б) Преобразование сферических координат



$$R_x(\epsilon) = R_x^{-1}(-\epsilon) = R_x^T(\epsilon)$$

поворот на угол ϵ вокруг оси x

Сферические координаты λ и β являются углами в сферических координатах

Наћи небесне екваторске координате об. еклиптичког пола

$$\beta = 90^\circ, \epsilon \approx 23.5^\circ$$

$$\left. \begin{aligned} x_e &= \cos \beta \cos \lambda \\ y_e &= \cos \beta \sin \lambda \\ z_e &= \sin \beta = 1 \end{aligned} \right\} \Rightarrow (x_e, y_e, z_e) = (0, 0, 1)$$

$$\left. \begin{aligned} x &= x_e \\ y &= y_e \cos \epsilon - z_e \sin \epsilon \\ z &= z_e \sin \epsilon + x_e \cos \epsilon \end{aligned} \right\} \Rightarrow \begin{aligned} x &= 0 \\ y &= -\sin \epsilon \\ z &= \cos \epsilon \end{aligned}$$

$$(\alpha, \delta) = (270^\circ, 66.5^\circ)$$

Познато нам је и:

$$\begin{aligned} x &= \cos \delta \cos \alpha \\ y &= \cos \delta \sin \alpha \\ z &= \sin \delta \end{aligned}$$

$$\begin{aligned} \Rightarrow \cos \epsilon &= \sin \delta \\ \cos \epsilon &= \cos(90^\circ - \delta) \\ \delta &= 90^\circ - \epsilon = 66.5^\circ \end{aligned}$$

$$\begin{aligned} \Rightarrow \cos \delta \sin \alpha &= -\sin \epsilon \\ \sin \alpha &= -\frac{\sin \epsilon}{\cos \delta} = -\frac{\sin \epsilon}{\cos(90^\circ - \epsilon)} = -\frac{\sin \epsilon}{\sin \epsilon} = -1 \\ \Rightarrow \alpha &= 270^\circ \end{aligned}$$

Јако су об. координате: $\alpha = 63^\circ, \delta = 17^\circ$. Наћи λ и β . Проверити да ли се налазе на једној еклиптици или еклиптику?

$$\begin{aligned} x &= \cos \delta \cos \alpha = 0.4342 \\ y &= \cos \delta \sin \alpha = 0.8979 \\ z &= \sin \delta = 0.2923 \end{aligned}$$

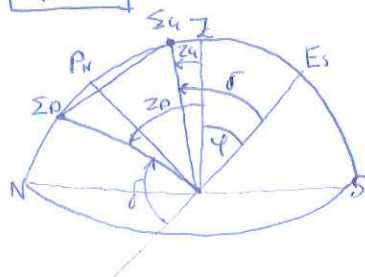
$$\begin{aligned} x_e &= x = 0.4342 \\ y_e &= \cos \epsilon y + \sin \epsilon z = 0.8979 \\ z_e &= -y \sin \epsilon + z \cos \epsilon = 0.0716 \end{aligned}$$

$$\begin{aligned} \Rightarrow \tan \lambda &= \frac{y_e}{x_e} \Rightarrow \lambda \approx 64.2^\circ \\ \sin \beta &= z_e \Rightarrow \beta \approx 4.1^\circ \\ |\beta| < |\lambda| &- \text{Јако се налазе на једној еклиптици} \end{aligned}$$

Iz Beograda ($\varphi = 44^{\circ}40'13''.2$) posmatramo je cirkumpolarnu zvezdu u obliku jednokrakog trougla.
 $Z_4' = 17^{\circ}18'40''.2$ i $Z_0' = 73^{\circ}1'11''.6$. Pretpostavljamo da su atmosfere uslovi isti za ove trenutke.
 (sećamo se od zenita)

- a) Odrediti vrednost k u $p = k \tan Z'$
 b) Odrediti A i B u razliku $p = A \tan Z' - B \tan^3 Z'$ formula δ iz prethodnog dela

a) $Z_4 = Z_4' + k \tan Z_4'$
 $Z_0 = Z_0' + k \tan Z_0'$
 $Z_4 = \varphi - \delta$ - izmereno od zenita
 $Z_0 = \delta - \varphi$ - izmereno od zenita



$$\delta + \varphi + Z_0 = 180^{\circ}$$

$$Z_0 = 180^{\circ} - \delta - \varphi$$

$$Z_0 + Z_4 = 180^{\circ} - 2\varphi$$

$$\varphi = \frac{180^{\circ} - (Z_0 + Z_4)}{2}$$

$$Z_4' + k \tan Z_4' + Z_0' + k \tan Z_0' = 180^{\circ} - 2\varphi$$

$$Z_4' + Z_0' + k (\tan Z_4' + \tan Z_0') = 180^{\circ} - 2\varphi$$

$$Z_4' = 17^{\circ}.3133$$

$$Z_0' = 73^{\circ}.0189$$

$$\varphi = 44^{\circ}.8036$$

$$\delta = Z_4 + \varphi = Z_4' + k \tan Z_4' + \varphi = 62^{\circ}7'15''.41$$

$$\Rightarrow k = \frac{180^{\circ} - 2\varphi - (Z_4' + Z_0')}{\tan Z_4' + \tan Z_0'}$$

$$k = 0^{\circ}.01655 = 56''.58$$

$$0.3117A - 0.0303B = 8''.8$$

$$3.27A - 35.12B = 155''.0$$

b) $p = A \tan Z_4' - B \tan^3 Z_4'$
 $p_0 = A \tan Z_0' - B \tan^3 Z_0'$

$$Z_4 = \delta - \varphi = Z_4' + A \tan Z_4' - B \tan^3 Z_4'$$

$$Z_0 = 180^{\circ} - \delta - \varphi = Z_0' + A \tan Z_0' - B \tan^3 Z_0'$$

$$\Rightarrow \delta - \varphi = Z_4' + A \tan Z_4' - B \tan^3 Z_4'$$

$$180^{\circ} - \delta - \varphi = Z_0' + A \tan Z_0' - B \tan^3 Z_0' \Rightarrow \begin{matrix} A = 60''.3 \\ B = 0''.069 \end{matrix}$$

Vrednost tačvost ne može se postići jer je u pitanju jedna cirkumpolarna zvezda

Izračunati, za posmatrača na $\varphi = 41^{\circ}22'30''$ iznos refrakcije u trenutku najviše deklinacije zv. čija je deklinacija $\delta = 64^{\circ}15'20''$.

- a) Za koliko ova zvezda dopire za taj posmatrač, uzimajući refrakciju, ranije (odnosno kasnije) u najveću deklinaciju
 b) Izračunati pravenu kažu u q ponoviti refrakciju u ND

a) $\delta = 64^{\circ}.2222$
 $\varphi = 41^{\circ}.375$

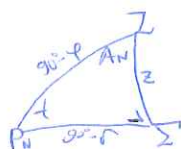
$$\cos(\delta_0 - \varphi) = \sin \delta \sin(\delta_0 - 2)$$

$$\sin \varphi = \sin \delta \cos Z$$

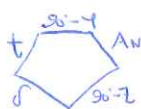
$$\cos Z = \sin \varphi \csc \delta$$

$$p = k \tan Z = p^*$$

dakle, ugađanje je naći Z preko φ i δ ;
 onda je p približna vrednost $\rightarrow$ razlika samo po sebi je približna,
 i mi stavljamo pravi zenitnu daljinu, a ne ugađanje (uže dato Z')



$$\varphi \leq \delta - \text{uslov za ND}$$

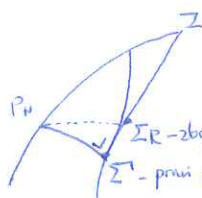


$$Z' = Z - p = Z - \frac{k \tan Z}{p^*}$$

$$Z = 42^{\circ}.7747 = 42^{\circ}46'28''.94$$

$$p^* = k \tan Z = 55''.65$$

$$Z' \approx 42^{\circ}.75924 = 42^{\circ}45'33''.27$$



ΣR - zbog refrakcije (u trenutku ND)

Σ' - pravi položaj

U zapadnoj neprekoj deklinaciji zvezdu posmatramo
 kašnje, tj. posle prođe kroz položaj ND (istočna strana obrnuto)
 zapad: pro se desi Σ pa tek onda ΣR

$$\cos \delta = \sin \varphi \csc Z$$

$$\tan \delta = \tan \varphi \sec Z$$

$$\frac{1}{\cos^2 \delta} d\delta = \frac{1}{\cos^2 Z} dZ \cdot \sec^2 Z$$

$$d\delta = \frac{\cos^2 Z}{\cos^2 \delta} \sec^2 Z dZ$$

$$\cos \delta = \tan \varphi \csc Z$$

$$\cos Z = \sin \varphi \csc \delta$$

$$d\delta = \frac{\tan^2 \varphi \csc^2 Z}{\sin^2 \varphi \csc^2 \delta} \sec^2 Z dZ$$

$$d\delta = \frac{\sin^2 \varphi \cdot \cos^2 \delta}{\cos^2 \varphi \cdot \sin^2 \delta} \cdot \frac{1}{\cos^2 \delta} dZ$$

$$d\delta = \cos \delta \sec^2 \varphi dZ$$

dZ distansi od refrakcije
 $\rightarrow$ onda imamo $d\delta$ uzal refrakcije

$$dZ = p = 55''.65$$

$$d\delta = 0.7723 \cdot 55''.65$$

$$d\delta = 42''.918 = 2''.87$$

Prava ND (uzal refrakcije)
 milisek na zapadu ranije,
 a na istom kasnije za 2''.87

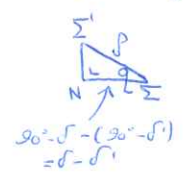
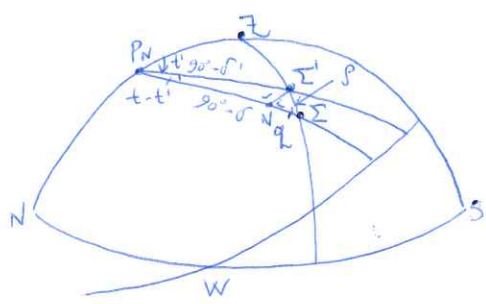
$$\begin{matrix} \delta = 15^{\circ} \\ \delta = 15' \\ \delta = 15'' \end{matrix}$$

b) Za promenu paralelnosti ugla usled refrakcije misao:

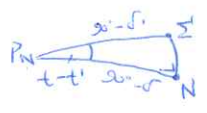
$dq = -tg \delta tg Z' \sin q' dz \rightarrow$ ubacuju se prave koordinate $\rightarrow$ Treba izvesti!

ND: $q = 90^\circ \rightarrow dq = -tg \delta tg Z dz = -106''.61 = -1'46''.61$

Izvesti ubicaj refrakcije na elipticne koordinate. Konstanti $p = k tg Z'$

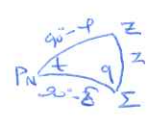


$N\Sigma' = \int \sin q = k tg Z' \sin q$
 $\delta' - \delta = N\Sigma = p \cos q = k tg Z' \cos q$



$\frac{\sin N\Sigma'}{\sin(t-t')} = \frac{\sin(90-\delta')}{\sin 90}$
 $\Rightarrow \sin N\Sigma' = \cos \delta' \sin(t-t')$
 $N\Sigma' = \cos \delta'(t-t')$

$\sin x \approx x$



$\delta' - \delta = k \frac{\sin Z'}{\cos Z'} \cos q$
 $t - t' = k \frac{\sin Z'}{\cos \delta'} \frac{\sin q}{\cos \delta'}$

$\cos Z = \sin p \sin \delta + \cos p \cos \delta \cos t$
 $\frac{\sin q}{\sin(90-\delta)} = \frac{\sin t}{\sin Z} \Rightarrow \sin Z \sin q = \sin t \cos p$
 $\sin Z \cos q = \cos(90-p) \sin(90-\delta) - \sin(90-p) \cos(90-\delta) \cos t$
 $\sin Z \cos q = \sin p \cos \delta - \cos p \sin \delta \cos t$

$Z' \approx Z$
 $\delta' - \delta = k \frac{\sin p \cos \delta - \cos p \sin \delta \cos t}{\sin p \sin \delta + \cos p \cos \delta \cos t}$
 $t - t' = k \frac{\sin t \cos p}{\cos \delta' (\sin p \sin \delta + \cos p \cos \delta \cos t)}$

Polazeći od Snellovog izraza za refrakciju $p = \frac{1}{m} \left[z' - \arcsin \left(\frac{\sin z'}{p_0^u} \right) \right]$ naći Bredlijev izraz:

$$p = \frac{2}{u} \frac{p_0^u - 1}{p_0^u + 1} \operatorname{tg} \left(z' - \frac{u p}{2} \right)$$

u - neki broj

p_0 - indeks refrakcije na Zem. površini

z' - ugaona zenitska duljina

Simpson: $r p^{u+1} = \text{const.}$
poluprodukt atm. sloja

$$p_u = z' - \arcsin \left(\frac{\sin z'}{p_0^u} \right)$$

$$\arcsin \left(\frac{\sin z'}{p_0^u} \right) = z' - p_u$$

$$\sin z' p_0^u = \sin (z' - p_u)$$

$$\frac{\sin z'}{\sin (z' - p_u)} = p_0^u$$

$$\text{Trik: } \frac{a}{b} = k \rightarrow \frac{a-b}{a+b} = \frac{k-1}{k+1}$$

$$\Rightarrow \frac{\sin z' - \sin (z' - p_u)}{\sin z' + \sin (z' - p_u)} = \frac{p_0^u - 1}{p_0^u + 1}$$

$$\Rightarrow \frac{2 \cos \left(z' - \frac{u p}{2} \right) \sin \frac{u p}{2}}{2 \sin \left(z' - \frac{u p}{2} \right) \cos \frac{u p}{2}} = \operatorname{tg} \frac{u p}{2} \operatorname{ctg} \left(z' - \frac{u p}{2} \right) = \frac{p_0^u - 1}{p_0^u + 1}$$

$$\operatorname{tg} \frac{u p}{2} = \frac{p_0^u - 1}{p_0^u + 1} \operatorname{tg} \left(z' - \frac{u p}{2} \right)$$

$$\left. \begin{array}{l} \sin \frac{p_u}{2} \approx \frac{p_u}{2} \\ \cos \frac{p_u}{2} \approx 1 \end{array} \right\} \Rightarrow \operatorname{tg} \frac{p_u}{2} \approx \frac{p_u}{2}$$

$$\Rightarrow \frac{p_u}{2} = \frac{p_0^u - 1}{p_0^u + 1} \operatorname{tg} \left(z' - \frac{u p}{2} \right)$$

$$p = \frac{2}{u} \frac{p_0^u - 1}{p_0^u + 1} \operatorname{tg} \left(z' - \frac{u p}{2} \right)$$

Koristimo:

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

Ako se posmatranjima ne može postići tačnost veća od 0.1 (tj. ako se ova veličina mora zanemariti) do koje zenitske duljine se možemo konstanti izrazom $p = k \operatorname{tg} z'$?

Opšti izraz za homogene udove u atmosferi: $p = 60''.3 \operatorname{tg} z' - 0''.063 \operatorname{tg}^3 z'$

Dobije: $0''.063 \operatorname{tg}^3 z' \leq 0.1$

$$\operatorname{tg}^3 z' \leq 1.5873$$

$z' \leq 49^\circ 23' 41''.19$ - za ove zenitske duljine vrijednost drugog člana ispod 0.1!

Pretpostavljajući da se Zemljina atmosfera općijeća homogeneu slojem atmosfere, visine H , polupradi:

$$p = \arcsin \left[\frac{R_\oplus}{R_\oplus + H} \sin z' (p^2 - 2p \cos \beta + 1)^{1/2} \right]$$

$$(p_1 \sin \alpha_1 = p_2 \sin \alpha_2)$$

$$OM = R_\oplus, NM = H$$

$$\sin i = p \sin r$$

$$i = \beta + r$$

$$\sin(\beta + r) = p \sin r$$

$$\sin \beta \cos r + \cos \beta \sin r = p \sin r$$

$$\sin \beta \cos r = (p - \cos \beta) \sin r$$

$$\sin^2 \beta \cos^2 r = (p^2 - 2p \cos \beta + \cos^2 \beta) \sin^2 r$$

$$\sin^2 \beta (1 - \sin^2 r) = (p^2 - 2p \cos \beta + \cos^2 \beta) \sin^2 r$$

$$\sin^2 \beta = (p^2 - 2p \cos \beta + 1) \sin^2 r$$

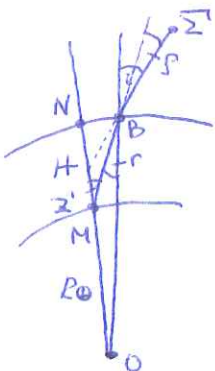
$$\sin \beta = (p^2 - 2p \cos \beta + 1)^{1/2} \sin r$$

$\triangle OMB$.

$$\frac{\sin(180^\circ - z')}{R_\oplus + H} = \frac{\sin r}{R_\oplus}$$

$$\sin r = \frac{R_\oplus}{R_\oplus + H} \sin z'$$

$$\sin \beta = (p^2 - 2p \cos \beta + 1)^{1/2} \frac{R_\oplus}{R_\oplus + H} \sin z'$$



U kojim slučajevima ~~deklinacija~~ nema uticaj na deklinaciju nekog tijela? $\rightarrow$ prebačeno u navedene yezbe

$$\cos(90^\circ - \delta) = \cos(90^\circ - \delta') \cos \beta + \sin(90^\circ - \delta') \sin \beta \cos(180^\circ - q)$$

$$\sin \delta = \sin \delta' \cos \beta - \cos \delta' \sin \beta \cos q$$

$$\delta = f(\beta)$$

$$\delta = f(0) + \beta f'(0) + \frac{\beta^2}{2} f''(0) + \dots$$

$$f(0) = \delta = \delta'$$

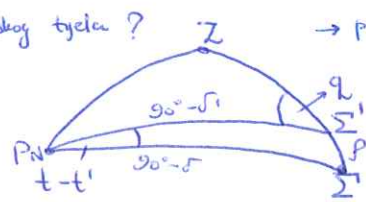
$$\cos \delta d\delta = -\sin \delta' \sin \beta d\beta - \cos \delta' \cos \beta \cos q d\beta$$

$$\cos \delta \frac{d\delta}{d\beta} = -\sin \delta' \sin \beta - \cos \delta' \cos \beta \cos q$$

$$f'(\beta) = \frac{d\delta}{d\beta} = \sec \delta (-\sin \beta' \sin \beta - \cos \delta' \cos \beta \cos q)$$

$$f'(0) = -\cos \delta' \cos q \sec \delta$$

$$f'(0) = -\cos q, \delta' = \delta$$



$$-\sin \delta \left(\frac{d\delta}{d\beta} \right)^2 + \cos \delta \frac{d^2 \delta}{d\beta^2} = -\sin \delta' \cos \beta + \cos \delta' \sin \beta \cos q$$

$$\sin \delta \left[1 - \left(\frac{d\delta}{d\beta} \right)^2 \right] = -\cos \delta \frac{d^2 \delta}{d\beta^2}$$

$$\Rightarrow \frac{d^2 \delta}{d\beta^2} = \left[\left(\frac{d\delta}{d\beta} \right)^2 - 1 \right] \tan \delta = f''(\beta)$$

$$f''(0) = (\cos^2 q - 1) \tan \delta = -\sin^2 q \tan \delta$$

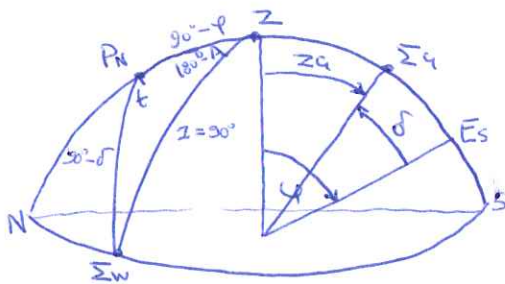
Dakle:

$$\delta = \delta' + \beta \cos q - \frac{\beta^2}{2} \sin^2 q \tan \delta + \dots$$

Ako se nši članovi zanemare: $\delta' - \delta = \beta \cos q$

$$\delta' - \delta = 0 \Rightarrow \cos q = 0 \Rightarrow q = \pm 90^\circ \rightarrow \text{ND}$$

Posmatrač je izmerio zenitsku daljinu zvezde Σ u trenutku njene CJK pismo od zenita i azimut u trenutku zalaza. Ne vodeći računa o refrakciji, naći φ . Kolika je greška ovako izračunate ~~refrakcije~~ geog. širine?



$$(1) \varphi = z_4 + \delta$$

$$z_4 \equiv z \Rightarrow z = \varphi - \delta, \delta = \varphi - z$$

$$(2) \sin \delta = -\cos \varphi \cos A$$

$$(1) \text{ i } (2) \Rightarrow \sin(\varphi - z) = -\cos \varphi \cos A$$

$$\sin \varphi \cos z - \cos \varphi \sin z = -\cos \varphi \cos A$$

$$\tan \varphi \cos z - \sin z = -\cos A$$

$$\tan \varphi = \frac{\sin z - \cos A}{\cos z} \Rightarrow \varphi = \arctan \left(\frac{\sin z - \cos A}{\cos z} \right)$$

$$d\varphi = \frac{1}{1 + \frac{(\sin z - \cos A)^2}{\cos^2 z}} \cdot \frac{\cos^2 z + (\sin z - \cos A) \sin z}{\cos^2 z} dz$$

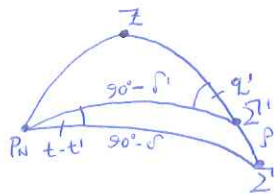
$$d\varphi = \frac{\cos^2 z}{\cos^2 z + \sin^2 z - 2 \sin z \cos A + \cos^2 A} \cdot \frac{\cos^2 z + \sin^2 z - \sin z \cos A}{\cos^2 z} dz$$

$$d\varphi = \frac{1 - \sin z \cos A}{1 - 2 \sin z \cos A + \cos^2 A} dz$$

$$dz = f = k \tan z'$$

$$z_4 \equiv z = z'$$

U kojim slučajevima refr. nema uticaja na δ nebeskog tela?



$q' \approx q$

$$\cos(90^\circ - \delta) = \cos(90^\circ - \delta') \cos \rho + \sin(90^\circ - \delta') \sin \rho \cos(180^\circ - q)$$

$$\sin \delta = \sin \delta' \cos \rho - \cos \delta' \sin \rho \cos q \quad (1)$$

$$\delta = f(\rho)$$

$$\delta = f(0) + \rho f'(0) + \frac{\rho^2}{2} f''(0) + \dots$$

$$f(0) = \delta = \delta'$$

analitička f-ja od ρ
na skali $\rho=0$
 $\delta(0) = \delta'$ - nepravilna refrakcija

δ' je izmjereni
visina!

$d(1) \rightarrow$

$$\cos \delta d\delta = -\sin \delta' \sin \rho d\rho - \cos \delta' \cos \rho \cos q d\rho$$

$$\cos \delta \frac{d\delta}{d\rho} = -\sin \delta' \sin \rho - \cos \delta' \cos \rho \cos q$$

$$f'(\delta) = \frac{d\delta}{d\rho} = \sec \delta (-\sin \delta' \sin \rho - \cos \delta' \cos \rho \cos q)$$

$$f'(0) = -\cos \delta' \cos q \sec \delta'$$

$$\delta' = \delta \Rightarrow f'(0) = -\cos q$$

$$-\sin \delta \left(\frac{d\delta}{d\rho} \right)^2 + \cos \delta \frac{d^2\delta}{d\rho^2} = -\sin \delta' \cos q + \cos \delta' \sin \rho \cos q$$

$$\sin \delta \left[1 - \left(\frac{d\delta}{d\rho} \right)^2 \right] = -\cos \delta' \frac{d^2\delta}{d\rho^2}$$

$$\frac{d^2\delta}{d\rho^2} = \left[\left(\frac{d\delta}{d\rho} \right)^2 - 1 \right] \tan \delta = f''(\delta)$$

za $2 < 70^\circ$
 $\delta = 1^\circ$ i $\delta' = 0$!

$$f''(0) = (\cos^2 q - 1) \tan \delta = -\sin^2 q \tan \delta$$

$$\text{Dakle: } \delta = \delta' - \rho \cos q - \frac{\rho^2}{2} \sin^2 q \tan \delta + \dots$$

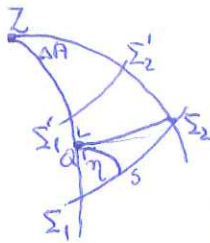
zanemarivanjem "viših" članova:

$$\cos q = 0 \Rightarrow q = 90^\circ \Rightarrow \text{ND}$$

Refrakcija puzanja zvezda ku
zemlji, a puzanja tog puzanja
ne deli. Knjig. dakle se sa $\cos q$

Dgstvo refrakcije je dato izrazom $\rho = k \tan z$. Pokazati da će njeno dgstvo na . daljina duge bliskih nebeskih (kompas-
duge zvezde) biti dato preko izraza:

$ds = k s (1 + \cos^2 \eta \tan^2 z)$, gde je s daljina zvezde (u sekundama ugla), a η ugao između Z.d. zapadne zvezde i luka s .



$\Delta Z_1 Z_2$:

$$\cos \delta = \cos z_1 \cos z_2 + \sin z_1 \sin z_2 \cos \Delta A$$

Refrakcija smanjuje z. daljine pa se smanjuje to s , a ΔA ostaje isto

$$-\sin \delta ds = -\sin z_1 \cos z_2 dz_1 - \cos z_1 \sin z_2 dz_2 + \cos z_1 \sin z_2 \cos \Delta A dz_1 + \sin z_1 \cos z_2 \cos \Delta A dz_2$$

$$-\sin \delta ds = dz_1 (-\sin z_1 \cos z_2 + \cos z_1 \sin z_2 \cos \Delta A) + dz_2 (-\cos z_1 \sin z_2 + \sin z_1 \cos z_2 \cos \Delta A)$$

ovde samo uvek
ima refrakciju
pa se zmanjuje
daljina sunnjuge

$$\left\{ \begin{aligned} dz_1 &= -k \tan z_1 \\ dz_2 &= -k \tan z_2 \\ \sin^2 \frac{\Delta A}{2} &= \frac{1 - \cos \Delta A}{2} \\ \cos \Delta A &= 1 - 2 \sin^2 \frac{\Delta A}{2} \end{aligned} \right\} \Rightarrow$$

$$-\sin \delta ds = k \tan z_1 \sin z_1 \cos z_2 - k \tan z_1 \cos z_1 \sin z_2 (1 - 2 \sin^2 \frac{\Delta A}{2}) + k \tan z_2 \cos z_1 \sin z_2 - k \tan z_2 \sin z_1 \cos z_2 (1 - 2 \sin^2 \frac{\Delta A}{2})$$

$$\Rightarrow -\sin \delta ds = k \sin z_1 \cos z_2 (\tan z_1 - \tan z_2) - k \cos z_1 \sin z_2 (\tan z_1 - \tan z_2) + 4k \sin z_1 \sin z_2 \sin^2 \frac{\Delta A}{2}$$

$$-\sin \delta ds = k (\tan z_1 - \tan z_2) (\sin z_1 \cos z_2 - \cos z_1 \sin z_2) + 4k \sin z_1 \sin z_2 \sin^2 \frac{\Delta A}{2}$$

$$-\sin \delta ds = k \left(\frac{\sin z_1}{\cos z_1} - \frac{\sin z_2}{\cos z_2} \right) \sin(z_1 - z_2) + 4k \sin z_1 \sin z_2 \sin^2 \frac{\Delta A}{2}$$

$$\frac{\sin z_1 \cos z_2 - \sin z_2 \cos z_1}{\cos z_1 \cos z_2} = \frac{\sin(z_1 - z_2)}{\cos z_1 \cos z_2}$$

$$\Rightarrow -\sin \delta ds = k \sin^2(z_1 - z_2) \sec z_1 \sec z_2 + 4k \sin z_1 \sin z_2 \sin^2 \frac{\Delta A}{2}$$

$$\Delta A \text{ mali ugao} \rightarrow \sin s \approx s, 4 \sin^2 \frac{\Delta A}{2} \approx 4 \left(\frac{\Delta A}{2} \right)^2 = (\Delta A)^2, \Sigma_1 Q = Z_1 - Z_2 = s \cos \eta$$

$$\Rightarrow -\sin \delta ds = k s^2 \cos^2 \eta \sec^2 z_1 \sec^2 z_2 + k (\Delta A)^2 \sin z_1 \sin z_2$$

$$Z_1 \approx Z_2 \rightarrow -s ds = k s^2 \cos^2 \eta \sec^2 z + k (\Delta A)^2 \sin^2 z$$

$$\Rightarrow -s ds = k s^2 \cos^2 \eta \sec^2 z + k s^2 \sin^2 \eta$$

$$-ds = k s (\cos^2 \eta \sec^2 z + \sin^2 \eta)$$

$$\sec^2 z = 1 + \tan^2 z$$

$$-ds = k s (\cos^2 \eta + \cos^2 \eta \tan^2 z + \sin^2 \eta)$$

$$-ds = k s (1 + \cos^2 \eta \tan^2 z)$$

Za ovakvo treba povećati ujerenu
ugloma daljinu duge bliskih zvezda!
da bi se dobila stvarna!



$$(\Delta A)^2 = s^2 \sin^2 \eta \cos^2 z \sec^2 z$$

Sa druge:

$$\frac{\sin \Delta A}{\sin \Delta Z_2} = \frac{\sin 90^\circ}{\sin z_2}$$

$$\sin \Delta Z_2 \approx \Delta Z_2 = s \sin \eta$$

$$\sin \Delta A \approx \Delta A$$

$$\Rightarrow \Delta A = s \sin \eta / \sin z_2$$

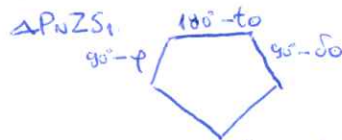
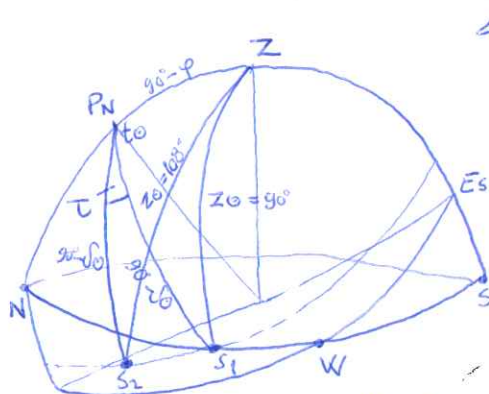
$$\Delta A = s \sin \eta \cos^2 z_2$$

Refrakcija prividno izdize nebeska tela. Takozvana posredna pruga nego što se on zaista događa, a zakaz nakon.
U zlatu: zakazu $Z = 90^\circ$. Za drugačiju refrakciju dajemo $Z = 90^\circ + \rho$. Metodu se služi Sunca obračunavamo i R0:

$$Z_0 = 90^\circ + \rho + R_0 = 90^\circ + 34' + 16' = 95^\circ 50'$$

refr. na horizontu!

Izračunati dužinu trajanja astronomske sumraka u Beogradu ($\varphi = 44^\circ 44' 13''$) kada je $\delta = 23^\circ 10'$. (građ. i naut.)



$$\varphi = 44.7367$$

$$\delta = 23.1667$$

$$\cos t_0 = -\tan \varphi \tan \delta \Rightarrow t_0 = 115.15 = 115.9'$$

$$\Delta PNZ_2: \cos(108^\circ) = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos(t_0 + t_2)$$

$$\cos(t_0 + t_2) = \frac{\cos(108^\circ) - \sin \varphi \sin \delta}{\cos \varphi \cos \delta}$$

oni mogu da ugore usne zvezke solsticijama

$$\cos(t_0 + t_2) = \frac{-0.586243}{0.652312}$$

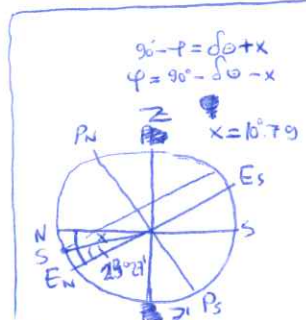
$$t_0 + t_2 = 153.98965$$

$$\Rightarrow t_2 = 2^h 35^m 21^s.72$$

$$\cos(t_2 + t_0) = \frac{\cos(96^\circ) - \sin \varphi \sin \delta}{\cos \varphi \cos \delta} \Rightarrow t_2 + t_0 = 125.319446 \Rightarrow t_2 = 0^h 42^m 40^s.67$$

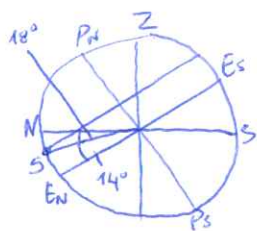
$$\cos(t_2 + t_0) = \frac{\cos(102^\circ) - \sin \varphi \sin \delta}{\cos \varphi \cos \delta} \Rightarrow t_2 + t_0 = 138.04931 \Rightarrow t_2 = 1^h 31^m 35^s.83$$

Pokazimo dugi intervali; posledicu velikog δ



a) Kolika je geografska širina mesta u kome astronomske sumraki traju celu noc drug datuma kada je $\delta = 14^\circ$?

b) Kolika je dužina astronomske sumraka u Moskvi ($\varphi = 55^\circ 45'$) ako je $\delta = 23^\circ 27'$?



$$ZEN = 90^\circ - \varphi$$

$$\Rightarrow 90^\circ - \varphi = 18^\circ + \delta$$

$$\varphi = 58^\circ$$

$$\delta > 90^\circ - \varphi - 18^\circ$$

$$14^\circ > 90^\circ - \varphi - 18^\circ$$

$$\varphi > 58^\circ$$

Pokazati da je za mesto na eliptiku trajanje ast. sumraka dato preko (u časovima):

$$T = \frac{12}{\pi} \arccos(\sin 18^\circ \sec \delta)$$

$$\Delta PNZ_1: \cos Z_0 = \sin \delta \cos 90^\circ + \cos \delta \sin 90^\circ \cos(T + 90^\circ)$$

$$Z_0 = 108^\circ$$

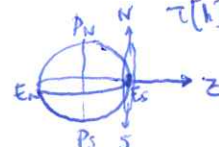
$$\cos 108^\circ = \cos \delta \cos(T + 90^\circ)$$

$$\cos 108^\circ = -\cos \delta \sin T$$

$$-\sin 18^\circ$$

$$\Rightarrow \sin 18^\circ = \cos \delta \sin T$$

$$T = \arcsin(\sin 18^\circ \sec \delta) \cdot \frac{1}{15} \cdot \frac{120^\circ}{\pi}$$



$$T_{\text{doba}}: \varphi = 0^\circ$$

$$\cos t_0 = -\tan \varphi \tan \delta \Rightarrow t_0 = 90^\circ$$

$$\cos(t_2 + t_0) = \frac{\cos 108^\circ}{\cos \delta}$$

$$-\sin t_2 = \frac{\cos 108^\circ}{\cos \delta}$$

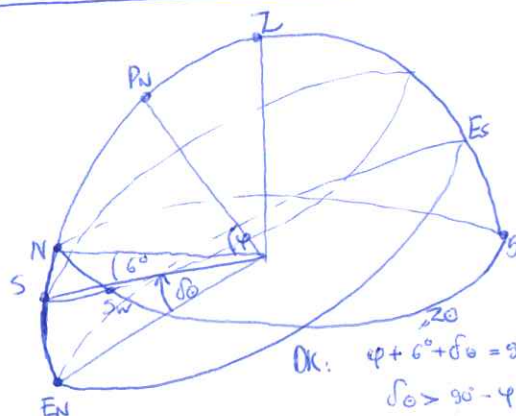
Teorijski uvod:

- Treptanje zvezda je posledica refrakcije.
- Sumraki su isto posledica refrakcije
- večernji sumrak (sunset) i jutarnji sumrak (solar)

Večernji građanski sumrak počinje od Sunčevog zalaza, a završava se kad Sunce dotigne 6° ispod horizonta. Jutarnji građanski sumrak počinje od kad Sunce dotigne usne od $h = -6^\circ$, pa do Sunčevog izlaza

Oko letnjeg solsticija, na $\varphi = 60^\circ$ s početkom večernjeg građanskog sumraka stapa se sa početkom jutarnjeg sumraka $\rightarrow$ građanski sumrak traje celu noc!

$\rightarrow$ to su bijele noci



$$\Delta K: \varphi + 6^\circ + \delta = 90^\circ$$

$$\delta > 90^\circ - \varphi - 6^\circ$$

$$\varphi = 60^\circ \Rightarrow \delta > 24^\circ$$

$$23.5^\circ > 84^\circ - \varphi$$

$$\varphi > 60.5^\circ$$

$$\delta_{\text{max}} = 23.5^\circ$$

Odrediti promene čas. ugla δ u toku godine, u fazi promene δ_0 . Rezultat primeniti na φ Beogradu, u dana selakija i elinogor.

$$\cos(90^\circ - \delta) = \sin \delta \sin \varphi + \cos \delta \cos \varphi \cos \alpha$$

$$0 = \cos \delta \sin \varphi d\delta + (\sin \delta \cos \varphi \cos \alpha) d\delta - \cos \delta \cos \varphi \sin \alpha d\alpha$$

$$\cos \delta \cos \varphi \sin \alpha d\alpha = d\delta (\cos \delta \sin \varphi - \sin \delta \cos \varphi \cos \alpha)$$

$$d\alpha = \left(\tan \varphi \frac{1}{\sin \delta} - \tan \delta \cot \varphi \right) d\delta$$

$$\left. \begin{array}{l} \varphi = 44^\circ 48' 13''.2 \\ \delta = 523^\circ.5 \\ \alpha = 0^\circ \end{array} \right\}$$

$$\cos \delta = -\tan \varphi \tan \delta$$

$$\delta_0 = 0^\circ$$

$$\cos \delta = 0 \Rightarrow \delta_0 = 90^\circ$$

$$d\alpha = \tan \varphi d\delta = 0.9886 d\delta$$

$$\delta = \pm 23.5^\circ$$

$$\cos \delta = -0.4289 \Rightarrow \delta = 115.21^\circ$$

$$d\alpha = 1.3 d\delta$$

do 31% veći uticaj na azimut u selakija nego elinogor!

U. Vemutlu $s = 6^h 42^m 53^s$ meridy. zv. vremena, refrakcyjna promena za zvezdu $\Sigma (\alpha = 3^h 5^m 26^s; \delta = 40^\circ 47.7')$ po paraleli i deklinacijama linije su jednake. Obrediti geograficu širinu mesta.

$$\delta' - \delta = k \frac{\sin \varphi \cos \delta - \cos \varphi \sin \delta \cos t}{\sin \delta \sin \varphi + \cos \delta \cos \varphi \cos t}$$

$$t - t' = \sec \delta \frac{\sin t \cos \varphi}{\sin \delta \sin \varphi + \cos \delta \cos \varphi \cos t}$$

$$\delta' - \delta = t - t' \Rightarrow \sin \varphi \cos \delta - \cos \varphi \sin \delta \cos t = \frac{\sin t \cos \varphi}{\cos \delta}$$

$$\sin \varphi \cos^2 \delta - \cos \varphi \sin \delta \cos t \cos \delta = \sin t \cos \varphi$$

$$\sin \varphi \cos^2 \delta - \cos \varphi (\sin \delta \cos \delta \cos t + \sin t) = 0$$

$$\tan \varphi = \frac{\sin \delta \cos \delta \cos t + \sin t}{\cos^2 \delta}$$

$$\varphi = 62^\circ 29' 55''.87$$

$$s = 6^h 42^m 53^s$$

$$\alpha = 3^h 5^m 26^s$$

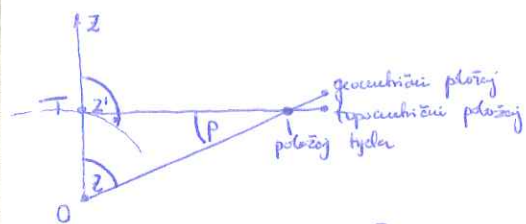
$$\delta = 40^\circ 47.7'$$

$$s = \alpha + t$$

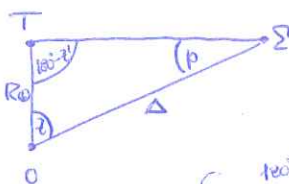
$$t = s - \alpha$$

$$t = 3^h 62^m 17^s = 54^\circ 36' 25''$$

Druhá paralela je mgro pod křivou se sa nekdy týče vlní Zemské plošce



$$\frac{\sin p}{\sin z'} = \frac{R_0}{\Delta} \Rightarrow p = \arcsin \left(\frac{R_0}{\Delta} \sin z' \right)$$



$$\frac{\sin p}{R_0} = \frac{\sin(180 - z')}{\Delta} \Rightarrow \frac{\sin p}{R_0} = \frac{\sin z'}{\Delta}$$

$$\left\{ \begin{array}{l} 180 - z' + p + z = 180 \\ \Rightarrow z' = z + p \\ \sin p = \frac{R_0}{\Delta} \sin z \\ z' = 90^\circ \Rightarrow p = \sin p \approx \frac{R_0}{\Delta} \\ \sin p \approx p \approx \frac{R_0}{\Delta} \sin z' \end{array} \right\} \Rightarrow p = \frac{R_0}{\Delta} \sin z'$$

• vliv dle druhé paralely na horizontální koordinátu

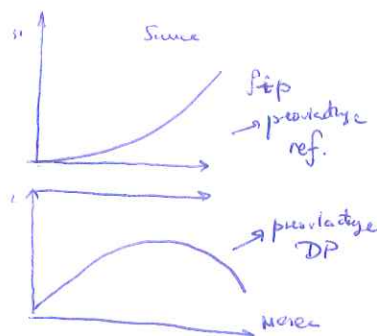
$$A' = A, z' = z + p = z + \frac{R_0}{\Delta} \sin z' \rightarrow DP \text{ přešlun zru. dle hru. sru. od refrakce}$$

Motivace, Zemské nje sféra, b. plošce nje se rozdílne u zru. od topocentru

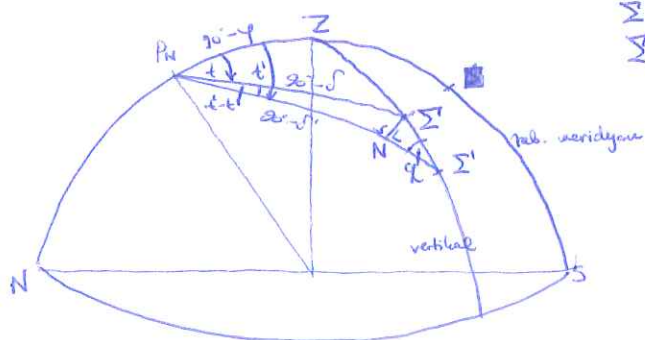
$$\sin p = \frac{R_0}{\Delta} \sin z'$$

$\Delta p_H \approx 6' 17''$ } rozdílne dle druhé paralely křiv se
 $\Delta p_B \approx 0' 03''$ } křiv referenční dle druhé uvažuje plavni i dle druhé plošce Země

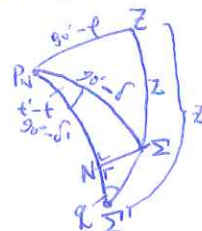
$$p = \frac{R_0}{\Delta \sin z'} - \text{druhá paralela} \Rightarrow \text{za prázdnou R: } p = \frac{R}{\Delta \sin z'} = \frac{R}{R_0 \Delta \sin z'} = \frac{R}{R_0} p_c$$



• vliv dle druhé paralely na dle druhé koordinátu:



Σ -první geocentrický plošný
 Σ' -první topocentrický plošný



$$\Delta N \Sigma \Sigma' \left\{ \begin{array}{l} N \Sigma = \Sigma' \sin p \\ \Sigma' = p = \frac{R_0}{\Delta} \sin z' \end{array} \right\} \cdot N \Sigma' = \frac{R_0}{\Delta} \sin z' \sin p$$

$$\Delta P \Sigma \Sigma' \left\{ \begin{array}{l} \frac{\sin N \Sigma}{\sin(t-t')} = \frac{\sin(90 - \delta)}{\sin 90} \end{array} \right.$$

$$\Rightarrow \frac{\sin N \Sigma}{\sin(t-t')} = \cos \delta \Rightarrow N \Sigma \approx \sin(t-t') \cos \delta \quad (2)$$

$$N \Sigma' = (90 - \delta') - (90 - \delta) = \delta - \delta' \quad (3)$$

$$(1) \text{ i } (3) \Rightarrow \delta - \delta' = \frac{R_0}{\Delta} \sin z' \cos \delta \quad (4)$$

$$(1) \text{ i } (2) \Rightarrow (t-t') \cos \delta = \frac{R_0}{\Delta} \sin z' \sin \delta \quad (5)$$

$$\Delta P \Sigma \Sigma'$$

$$\frac{\sin z'}{\sin t'} = \frac{\sin(90 - \delta)}{\sin \delta} \Rightarrow \sin z' \sin \delta = \cos \delta \sin t'$$

$$\sin z' \cos \delta = \cos(90 - \delta) \sin(90 - \delta') - \sin(90 - \delta) \cos(90 - \delta') \cos t'$$

$$\sin z' \cos \delta = \sin \delta \cos \delta' - \cos \delta \sin \delta' \cos t'$$

$$\cos \delta' \approx \cos \delta$$

$$\sin t' \approx \sin t$$

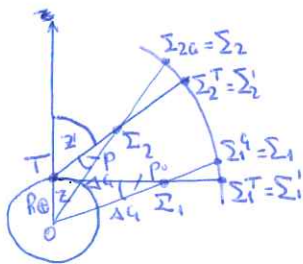
$$\left\{ \begin{array}{l} \delta' - \delta = -\frac{R_0}{\Delta} (\sin \delta \cos \delta' - \cos \delta \sin \delta' \cos t) \\ t' - t = \frac{R_0}{\Delta} \cos \delta \cos \delta' \sin t \end{array} \right.$$

p.-horizontale paralleliser

$$R_{\oplus} = 4.94 \text{ cm}$$

$$\sin \rho_0 = \frac{R_0}{\Delta G}$$

$$p_0 = \frac{P_0}{\Delta G_{\text{max}}}$$



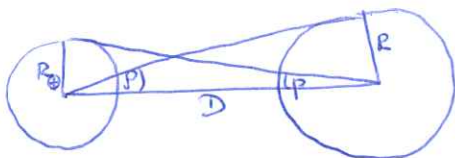
$$\frac{\sin P}{\sin P_0} = \frac{\sin Z'}{\sin \Delta_1}$$

$$\Rightarrow s_{up} = \frac{R_{\phi}}{\Delta} \sin \alpha$$

$$A' = A$$

— duenna
— parolulise
produs grăta
tyda ke homahe

Udretivače pravi veličina rešetkih tela u Sivičevom sistemu



ako zamo paralelnu p , odvisno deljine neb. kla

→ izmērīmu privādu poliprotēnu

$$\text{slup} = \frac{P_0}{D}$$

$$R_{\theta} = V_{scap}$$

$$R = D \sin \beta$$

$$\sup f = \frac{R}{D \mid R}$$

$$\frac{R}{R_0} = \frac{\sin \theta}{\sin \phi}$$

22

$$R \simeq R \oplus \frac{p}{10}$$

$$P_{\oplus} = 6378.13 \text{ km}$$

Решет. карт.

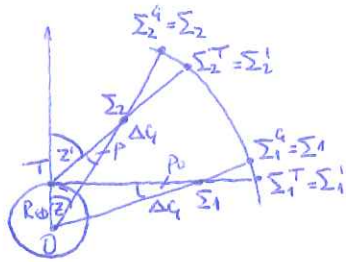
$$P = 15^{\circ} 32''.6$$

$$p = 57'2''.7$$

173⁸⁷ ~~4~~ km

$$R = 0.272 R_0 =$$

Dnevna paralela



$$z' = z + p$$

$$z = z'$$

dnevna paralela — ugao pod kojim se sa
nekej nekakvog tijela vidi R₀

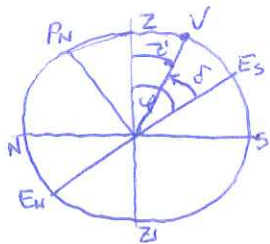
p — horizontalna paralela, z — topocentrična z. daljina, z' — geocentrična z. daljina

$$\Delta T \Sigma_1: \left. \begin{array}{l} \sin p = \frac{R_0}{\Delta_1} \\ \sin p \approx p \text{ "sin"} \end{array} \right\} \Rightarrow p = \frac{R_0}{\sin \Delta_1}$$

$z' = z + p$ — dnevna paralela
 $A' = A$ — pravični spita tijela
ku horizontu

$$\Delta O \Sigma_2: \left. \begin{array}{l} \frac{\sin p}{R_0} = \frac{\sin(180^\circ - z')}{\Delta_1} \\ \sin p = \frac{R_0}{\Delta_1} \sin z' \\ \sin p \approx p \text{ "sin"} \end{array} \right\} \Rightarrow \left. \begin{array}{l} p = \frac{R_0}{\sin \Delta_1} \sin z' \\ p = p \sin z' \end{array} \right\}$$

Uzmimo za geocentričnu daljinu Venere $\Delta_1 = 0.723 \text{ AJ}$, i da je njena deklinacija $\delta = 9^\circ 31'$, izračunati uticaj dnevne paralele na
horizontalne koordinate u trenutku GK Venere u Beogradu ($\varphi = 44^\circ 48' 13.2''$). Udalj R₀ = 6378.13 km



$$\varphi = 44^\circ 48' 13.2''$$

$$\delta = 9^\circ 31'$$

$$p = \frac{R_0}{\sin \Delta_1} = 12.163$$

$$1 \text{ AJ} = 149.597 \cdot 10^6 \text{ km}$$

$$z = z' - p = z' - p \sin z'$$

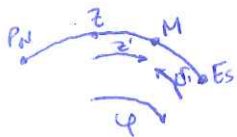
$$\left. \begin{array}{l} \Delta z = p \sin z' \\ z' = \varphi - \delta = 35^\circ 75' 36.7'' \end{array} \right\} \Rightarrow \left. \begin{array}{l} \Delta z = 7.107 \\ \Delta A = 0 \end{array} \right\}$$

U trenutku kada je Mjesecova horizontalna paralela Gta $p = 57' 13''$ postaviti su se uticaji DP i refrakcije. Kolika je Gta
topocentrična zenitna daljina Mjeseca u tom trenutku?

$$\left. \begin{array}{l} p = k \sin z' \\ p = p \sin z' \end{array} \right\} \Rightarrow p = p \Rightarrow k \sin z' = p \sin z'$$

$$\cos z' = \frac{k}{p} = 0.017573 \Rightarrow z' = 88^\circ 55' 35.102''$$

Horizontalna paralela Mjesec u trenutku njegove GK u Beogradu iznosi $p = 57' 13.3''$. Ako je njegova topocentrična deklinacija
 $\delta' = 22^\circ 17'$, uz R₀ = 6378.13 km, izračunati geocentričnu i topocentričnu rastojanja do Mjeseca.

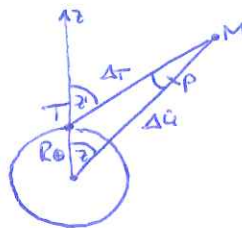


$$z' = \varphi - \delta'$$

$$p = p \sin z'$$

$$\Delta_1 = \frac{R_0}{\sin \varphi}$$

$$\sin p = \frac{R_0}{\Delta_1}$$



$$\frac{\sin p}{R_0} = \frac{\sin(180^\circ - z')}{\Delta_1}$$

$$\Delta_1 = \frac{R_0}{\sin p} = 376622.95 \text{ km}$$

$$\frac{\sin z'}{\Delta_1} = \frac{\sin z'}{\Delta_1}$$

$$\Delta T = \Delta_1 \frac{\sin z'}{\sin z'} = 370722.74 \text{ km}$$

$$z' = 22^\circ 52'$$

$$p = 22^\circ 29' = 22' 17.93'' = 0.371$$

$$z = z' - p = z' - p \sin z' = 22^\circ 14' 35''$$

Na koje rastojanje bi trebalo da se nalazi nekakvo tijelo da bi se uticaji refrakcije i paralele poništili na $z' = 60^\circ$?

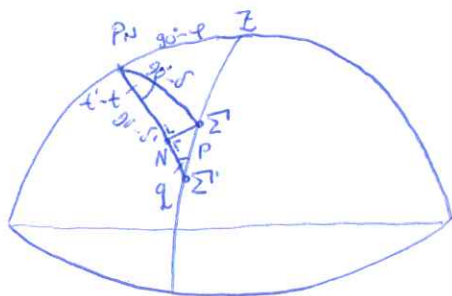
$$\delta = p$$

$$k \sin z' = p \sin z'$$

$$\frac{1}{\cos z'} k = p = \frac{R_0}{\sin \Delta}$$

$$\Delta = \frac{R_0 \cos z'}{k \sin z'} \Rightarrow \Delta = 10.94 \cdot 10^6 \text{ km}$$

Izvesti ubrzanje DP na danitoides koordinata i pokazati da je ubrzanje na δ jednako nuli u trenutku ND.



$$\begin{aligned} \Sigma \Sigma' &= p = p_0 \sin \delta' \\ N \Sigma &= \sin \varphi \Sigma \Sigma' = p_0 \sin \delta' \sin \varphi \\ N \Sigma' &= \cos \varphi \Sigma \Sigma' = p_0 \sin \delta' \cos \varphi \end{aligned} \quad \} (1)$$

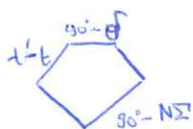
$$N \Sigma' = \delta - \delta' = p_0 \sin \delta' \cos \varphi$$

$$\sin \delta' \cos \varphi = \cos(90^\circ - \varphi) \sin(90^\circ - \delta') = \sin(90^\circ - \varphi) \cos(90^\circ - \delta') \cos \delta'$$

$$\sin \delta' \cos \varphi = \sin \varphi \cos \delta' - \cos \varphi \sin \delta' \cos \delta'$$

$$\delta' - \delta = -p_0 (\sin \varphi \cos \delta' - \cos \varphi \sin \delta' \cos \delta')$$

$$\rightarrow t \pm t' \Rightarrow \boxed{\Delta t = p_0 \sec \delta \cos \varphi \sin \delta}$$



$$\sin N \Sigma' = \cos \delta \sin(t' - t)$$

$$N \Sigma' \approx \cos \delta(t' - t)$$

$$\sin \delta' \sin \varphi = \cos \varphi \sin \delta'$$

$$\} \Rightarrow t' - t = \sec \delta p_0 \cos \varphi \sin \delta'$$

$$\Delta \delta = 0 \Rightarrow p_0 (\cos \varphi \sin \delta' \cos \delta' - \sin \varphi \cos \delta') = 0$$

$$p_0 \sin \delta' \cos \delta' = 0 \Rightarrow \boxed{q = 90^\circ} \text{ ND}$$

Horizontalna paralelna udjerenja varira u granicama $d\varphi = \pm 185''$. Uzimajući da je sr. daljina udjerenja od Zemlje $r = 389403 \text{ km}$, i da je $R_\oplus = 6378.13 \text{ km}$, naći: najpali i najudaljenije udjerenje.

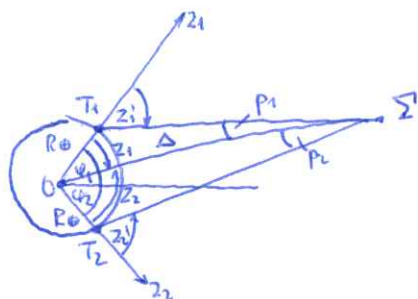
$$p_0 = \frac{R_\oplus}{\sin \delta' \Delta}$$

$$dp_0 = \left(\frac{R_\oplus}{\sin \delta'} \right) \left(-\frac{1}{\Delta^2} \right) d\Delta$$

$$d\Delta = -\frac{\Delta^2 \sin \delta' d\varphi}{R_\oplus}$$

$$d\Delta = \pm 22228.21 \text{ km}$$

Pod pretpostavom da je Zemlja loptica poluprečnika $R_\oplus$ odrediti geocentrične daljine tjeles koje u trenutku posmatranja daju posmatrača $T_1(\varphi_1)$; $T_2(\varphi_2)$, i njih ugaženje dugu Z_1, Z_2 .



$$Z_1 + Z_2 = \varphi_1 + \varphi_2$$

$$\begin{aligned} Z_1' &= p_1 + \varphi_1 \\ Z_2' &= p_2 + \varphi_2 \end{aligned} \quad \} (1) \Rightarrow Z_1' + Z_2' = p_1 + p_2 + \varphi_1 + \varphi_2$$

$$Z_1' + Z_2' = p_1 + p_2 + \varphi_1 + \varphi_2$$

$$\Rightarrow p_1 + p_2 = (Z_1' + Z_2') - (\varphi_1 + \varphi_2) \equiv p$$

$$\Delta OT_1 \Sigma' \quad \frac{\sin p_1}{R_\oplus} = \frac{\sin Z_1'}{\Delta}$$

$$\Delta OT_2 \Sigma \quad \frac{\sin p_2}{R_\oplus} = \frac{\sin Z_2'}{\Delta}$$

$$\sin p_2 = \frac{R_\oplus}{\Delta} \sin Z_2'$$

$$\sin p_1 = \frac{R_\oplus}{\Delta} \sin Z_1'$$

$$\sin \underbrace{(p - p_1)}_{p_2} = \sin p \cos p_1 - \cos p \sin p_1 = \frac{R_\oplus}{\Delta} \sin Z_2'$$

$$\sin p \cos p_1 - \frac{R_\oplus}{\Delta} \sin Z_1' \cos p = \frac{R_\oplus}{\Delta} \sin Z_2'$$

$$\cos p_1 = \frac{R_\oplus}{\Delta} \sec p (\sin Z_1' \cos p + \sin Z_2')$$

$$\cos p_1 = \sec p \frac{R_\oplus}{\Delta} (\sin Z_2' + \sin Z_1' \cos p) // \sin p_1 \quad \sin p_1 = \frac{\sin Z_1'}{\Delta} R_\oplus$$

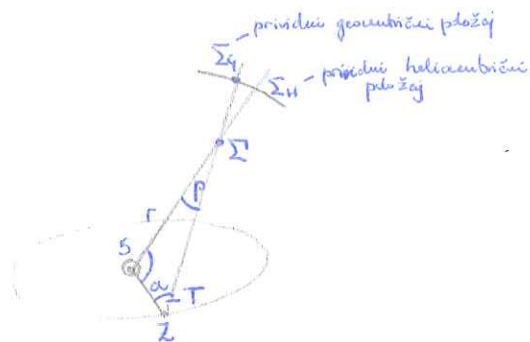
$$p = (Z_1' + Z_2') - (\varphi_1 + \varphi_2)$$

$$d\varphi p_1 = \sec p \sec Z_1' (\sin Z_1' + \sin Z_2' \cos p)$$

$$d\varphi p_1 = \frac{\sin Z_1' + \sin Z_2' \cos p}{\sin p \sin Z_1'} \rightarrow p_1 \rightarrow \Delta$$

Gedőjara paralaksza

Bessel 1838. és Lygni
Heidelberg 1879. α Cen



ΔΣΣE:

$$\frac{\sin p}{a} = \frac{\sin T}{r}$$

$$\sin p = \frac{a}{r} \sin T$$

$$\sin p = P \sin T$$

$$\Rightarrow P = \frac{a}{r \sin T} \sin T$$

π - gedőjara paralaksza

$$P = \pi \sin T$$

úgy is izmétel geocentrikus és heliocentrikus pályájára zvezde

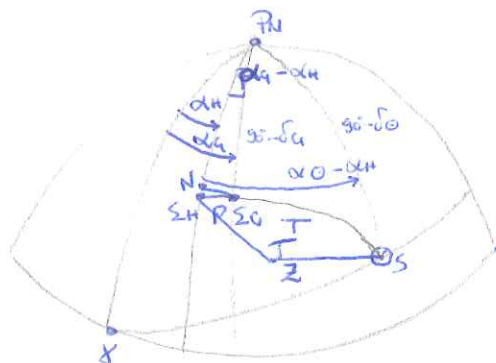
T = 90° → P = π - gedőjara paralaksza je nagyobb ugyan pol
közben se ulti vetülete párhuzamos Zvezde heliocentrikus orbite

Izračati uticaj gedőjara paralaksze na dvavtorisne koordinete

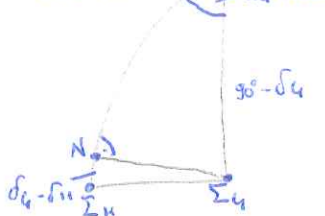
Geocentrikus pályájára zvezde natrzi se u
obrazu heliocentrikus pályájára, i to
poziciju kao Sencu!

ΔNΣHΣG

$$\left. \begin{aligned} \Sigma_G N &= \Sigma_G \Sigma_H \sin q \\ \Sigma_H N &= \Sigma_G \Sigma_H \cos q \\ \Sigma_H \Sigma_G &= P = \pi \sin T \end{aligned} \right\} \Rightarrow \begin{aligned} \Sigma_G N &= \pi \sin T \sin q \\ \Sigma_H N &= \pi \sin T \cos q \end{aligned}$$



$$90^\circ - \delta_H - (90^\circ - \delta_G) = \delta_G - \delta_H$$



$$\sin N \Sigma_G \approx N \Sigma_G$$

$$\sin(\alpha_G - \alpha_H) \approx \alpha_G - \alpha_H$$

ΔPNΣGΣH

$$\frac{\sin N \Sigma_G}{\sin(\alpha_G - \alpha_H)} = \frac{\sin(90^\circ - \delta_G)}{\sin 90^\circ} = \cos \delta_G$$

$$\sin N \Sigma_G = \sin(\alpha_G - \alpha_H) \cos \delta_G$$

$$N \Sigma_G = \sin(\alpha_G - \alpha_H) \cos \delta_G$$

$$N \Sigma_H = (\alpha_G - \alpha_H) \cos \delta_G$$

$$N \Sigma_H = \delta_G - \delta_H$$

ΔPNΣHΣG

$$\frac{\sin T}{\sin(\alpha_G - \alpha_H)} = \frac{\sin(90^\circ - \delta_G)}{\sin q}$$

$$\sin T \sin q = \cos \delta_G \sin(\alpha_G - \alpha_H)$$

$$\sin T \cos q = \cos(90^\circ - \delta_G) \sin(90^\circ - \delta_H) - \sin(90^\circ - \delta_G) \cos(90^\circ - \delta_H) \cos(\alpha_G - \alpha_H)$$

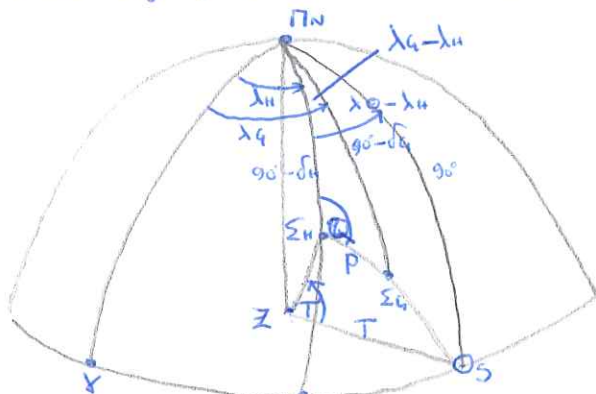
$$\sin T \cos q = \sin \delta_G \cos \delta_H - \cos \delta_G \sin \delta_H \cos(\alpha_G - \alpha_H)$$

$$(4) \rightarrow (-) \Rightarrow$$

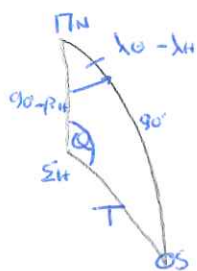
$$\left. \begin{aligned} \alpha_G - \alpha_H &= \pi \sin(\alpha_G - \alpha_H) \cos \delta_G \sec \delta_G \\ \delta_G - \delta_H &= \pi [\sin \delta_G \cos \delta_H - \cos \delta_G \sin \delta_H \cos(\alpha_G - \alpha_H)] \end{aligned} \right\}$$

Uz poznavanje paralaksze, može se izračiti
prelaz na heliocentrične na geocentrične koordinate
i obrnuto!

Izračun abscise gledišne paralakse na eliptičke koordinate



$\Delta \Pi N \Sigma_H \Sigma_S$



$$\frac{\sin T}{\sin(\lambda_S - \lambda_H)} = \frac{\sin 90^\circ}{\sin Q}$$

$$\sin T \sin Q = \sin(\lambda_S - \lambda_H)$$

$$\boxed{\lambda_S - \lambda_H = \pi \sin(\lambda_S - \lambda_H) \sin \beta_H}$$

$\Delta \Pi N \Sigma_H \Sigma_S$

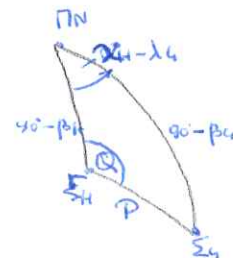
$$\frac{\sin(\lambda_S - \lambda_H)}{\sin P} = \frac{\sin Q}{\sin(90^\circ - \beta_H)}$$

$$\sin(\lambda_S - \lambda_H) \cos \beta_H = \sin Q \sin P$$

$$\beta_S \approx \beta_H$$

$$(\lambda_S - \lambda_H) \cos \beta_H = \sin Q \sin P$$

$$(\lambda_S - \lambda_H) \cos \beta_H = \pi \sin T \sin Q$$



$\Delta \Pi N \Sigma_H \Sigma_S$

$$\sin T \cos Q = \cos 90^\circ \sin(90^\circ - \beta_H)$$

$$- \sin 90^\circ \cos(90^\circ - \beta_H) \cos(\lambda_S - \lambda_H)$$

$$\sin T \cos Q = - \sin \beta_H \cos(\lambda_S - \lambda_H)$$

$$\boxed{\beta_S - \beta_H = - \pi \cos(\lambda_S - \lambda_H) \sin \beta_H}$$

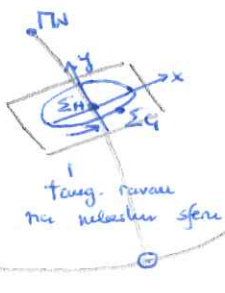
$\Delta \Sigma_S \Sigma_H \Pi_N$

$$\sin P \cos Q = \cos(90^\circ - \beta_S) \sin(90^\circ - \beta_H) - \sin(90^\circ - \beta_S) \cos(90^\circ - \beta_H) \cos(\lambda_H - \lambda_S)$$

$$\sin P \cos Q = \sin \beta_S \cos \beta_H - \cos \beta_S \sin \beta_H \cos(\lambda_H - \lambda_S)$$

$$\cos(\lambda_H - \lambda_S) \approx 1 \Rightarrow \frac{\sin P \cos Q}{\approx P} \approx \beta_S - \beta_H$$

$$\Rightarrow \beta_S - \beta_H = P \cos Q = \pi \sin T \cos Q$$



Geometrijsko mjesto geocentričnih položaja zvezde prema heliocentričnom u toku godine

$$\text{Konjunktura } \Sigma \text{ i } \odot (\lambda_H = \lambda_S)$$

$$\text{Opozicija } \Sigma \text{ i } \odot (\lambda_H = \lambda_S + 180^\circ)$$

$$\text{Kvadratura } \Sigma \text{ i } \odot (\lambda_H = \lambda_S + 90^\circ, \lambda_H = \lambda_S + 270^\circ)$$

U slučaju GP:

$$\Delta \lambda = 0$$

$$\Delta \beta \propto \pi \sin \beta_H$$

$$\Delta \lambda \propto \pi \sec \beta_H$$

$$\Delta \beta = 0$$

Dakle, heliocentrični položaj je nepromjenjiv, a geocentrični se mijenja u toku godine

$$x = (\lambda_S - \lambda_H) \cdot \cos \beta_H = \pi \sin(\lambda_S - \lambda_H)$$

$$y = \beta_S - \beta_H = - \pi \cos(\lambda_S - \lambda_H) \sin \beta_H$$

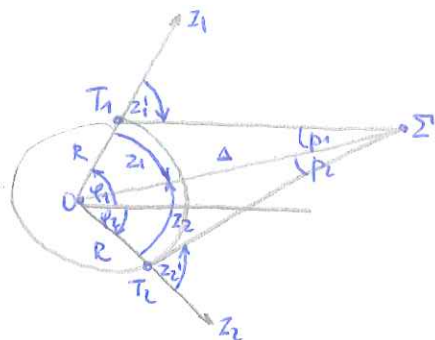
-treba eliminirati vreme
kako se kreće u λ_S

$$\frac{x}{\pi} = \sin(\lambda_S - \lambda_H) \quad \left/ \begin{matrix} \cos^2 \beta_H \\ \sin^2 \beta_H \end{matrix} \right. + \Rightarrow \frac{x^2}{\pi^2} + \frac{y^2}{\pi^2 \sin^2 \beta_H} = 1$$

→ geocentrični položaj u toku godine oko heliocentričnog opjeva elipse!

Paralaktičke elipse su projekcije Zemljine putanje oko Sunca na nebesnu sferu → egzaktan dokaz heliocentrizma

Pod pretpostavkom da je Zemlja loptu poluprečnika R , odrediš geocentričnu dužinu tela koje u istom trenutku posmatraju dva posmatrača $T_1(\varphi_1), T_2(\varphi_2)$ i iz svojih merenja daju z_1', z_2'



$$\begin{aligned} z_1 + z_2 &= \varphi_1 + \varphi_2 \\ z_1' &= z_1 + p_1 \\ z_2' &= z_2 + p_2 \\ z_1' + z_2' &= (z_1 + z_2) + p_1 + p_2 \\ p_1 + p_2 &= (z_1' + z_2') - (\varphi_1 + \varphi_2) = p \end{aligned}$$

$$\begin{aligned} \Delta OT_1 \Sigma: \quad \frac{\sin p_1}{R} &= \frac{\sin z_1'}{\Delta} \\ \Delta OT_2 \Sigma: \quad \frac{\sin p_2}{R} &= \frac{\sin z_2'}{\Delta} \end{aligned}$$

$$\begin{aligned} \sin p_2 &= \frac{R}{\Delta} \sin z_2' \\ \sin(p - p_1) &= \sin p \cos p_1 - \cos p \sin p_1 = \frac{R}{\Delta} \sin z_1' \end{aligned}$$

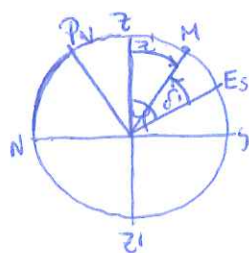
$$\begin{aligned} \sin p \cos p_1 - \frac{R}{\Delta} \sin z_1' \cos p &= \frac{R}{\Delta} \sin z_2' \\ \cos p_1 &= \cos p \cdot \frac{R}{\Delta} (\sin z_2' + \sin z_1' \cos p) / \sin p_1 \\ \cot p_1 &= \cos p \cdot \frac{1}{\frac{R}{\Delta} \sin z_1'} \cdot \frac{R}{\Delta} (\sin z_2' + \sin z_1' \cos p) \\ \cot p_1 &= \cos p \cdot \csc z_1' (\sin z_2' + \sin z_1' \cos p) \\ \cot p_1 &= \frac{\sin z_2' + \sin z_1' \cos p}{\sin p \sin z_1'} \rightarrow p_1 \rightarrow \Delta \end{aligned}$$

Asteroid prolazi pored Zemlje na rastojanju od 200000 km, $R_\oplus = 6378$ km

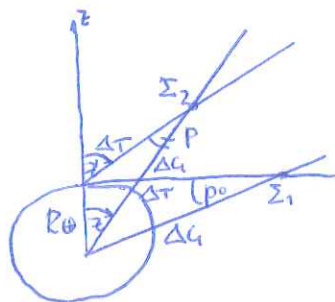
a) Kolika je njegov horizontalna paralaksa?

$$p_0 = \frac{R_\oplus}{\Delta \sin 1''} = 1''.827$$

b) On prolazi kroz meridijan mesta širine $\varphi = 56^\circ 20'$ na $z' = 35^\circ$. Kolika je njegov geocentrični doblinac?



$$\begin{aligned} \varphi &= z' + \delta' \\ \delta' &= \varphi - z' = 21^\circ 20' \\ z &= z' - p = z' - p_0 \sin z' = 33^\circ.952 \\ \Rightarrow \delta &= \varphi - z = 22^\circ.381 \end{aligned}$$



$$\frac{\sin p}{R_\oplus} = \frac{\sin z'}{\Delta_4}$$

$$\sin p = \frac{R_\oplus}{\Delta_4} \sin z'$$

$$p = \frac{R_\oplus}{\Delta_4 \sin 1''} \sin z' = p_0 \sin z'$$

$$z' = z + p$$

$$\sin p_0 = \frac{R_\oplus}{\Delta_4}$$

$$p_0 = \frac{R_\oplus}{\Delta_4 \sin 1''}$$

Odstrovanje pravek relativne neb. tela u S. sistemu



$$\sin \theta = \frac{R_{\odot}}{D}$$

$$\sin \phi = \frac{R}{D}$$

$$R_{\odot} = D \sin \theta$$

$$R = D \sin \phi$$

$$\frac{R}{R_{\odot}} = \frac{\sin \phi}{\sin \theta}$$

$$\frac{R}{R_{\odot}} \approx \frac{\phi}{\theta}$$

$$R = R_{\odot} \frac{\phi}{\theta}$$

Hiperh 2 v.p.n.e $\rightarrow p_H$

Francis 14.v. Parnu is Parnu i Kefene (Francuska Ligejuna)

$$T_H, R_H \rightarrow 36 \text{ z} \rightarrow R_{\odot}$$

$$R_{\odot} = 6378.13 \text{ km}$$

$$\text{Merce: } \phi = 15' 32''.6$$

$$\theta = 57' 2''.7$$

$$\Rightarrow R = 0.272 R_{\odot} = 1734.67 \text{ km}$$

$$p_0 \approx 8''.8$$

Dokazati da je za datu zvezdu Σ razlika između geometrijske, heliocentrične α i paralaksne α' jednaka je:

$$\lambda\theta = \arctan(-\cot\alpha \cos\epsilon)$$

$$\begin{aligned}\alpha' &= \alpha + \lambda\theta \\ \alpha &= \alpha_n \\ \delta' &= \delta_n \\ \delta &= \delta_n\end{aligned}$$

$$\alpha' - \alpha = \pi \sin(\alpha - \alpha_0) \cos\delta \sec\delta'$$

$$\alpha' - \alpha = -\pi \sin(\alpha - \alpha_0) \cos\delta \sec\delta'$$

$$= -\pi \sec\delta' (\sin\alpha \cos\lambda\theta \cos\delta \sec\delta' - \cos\alpha \sin\lambda\theta \cos\delta \sec\delta')$$

$$\cos\lambda\theta = \cos\alpha \cos\delta \sec\delta', \quad \sin\lambda\theta = \cot\epsilon \cot\delta \sec\delta', \quad \sin\delta \sec\delta' = \sin\epsilon \sin\lambda\theta$$

$$\sin\alpha \cos\delta \sec\delta' = \cot\epsilon \cot\delta \sec\delta' \cos\delta \sec\delta' = \cot\epsilon \sin\delta \sec\delta'$$

$$\sin\alpha \cos\delta \sec\delta' = \cot\epsilon \sin\epsilon \sin\lambda\theta = \cos\epsilon \sin\lambda\theta$$

$$\alpha' - \alpha = -\pi \sec\delta' (\sin\alpha \cos\lambda\theta - \cos\alpha \cos\epsilon \sin\lambda\theta)$$

$$\frac{d(\alpha' - \alpha)}{d\lambda\theta} = -\pi \sec\delta' (-\sin\alpha \sin\lambda\theta - \cos\alpha \cos\epsilon \cos\lambda\theta) = 0$$

$$\Rightarrow \sin\alpha \sin\lambda\theta = -\cos\alpha \cos\epsilon \cos\lambda\theta \Rightarrow \tan\lambda\theta = -\cot\alpha \cos\epsilon$$

Alto je uticaj GP na deklinaciju neke zvezde najveci u trenutku prolaska ravnodnevnice, pokazati da važi: $\tan\epsilon = \sin\alpha \tan\delta'$

$$\delta' - \delta = \pi (\sin\delta \cos\delta' - \cos\delta \sin\delta' \cos(\alpha - \alpha_0))$$

$$\delta' - \delta = \pi (\sin\delta \cos\delta' - \cos\delta \sin\delta' \cos\alpha \cos\epsilon + \cos\delta \sin\delta' \sin\alpha \sin\epsilon)$$

$$\delta' - \delta = \pi (\sin\epsilon \sin\lambda\theta \cos\delta' - \cos\delta \sin\delta' \cos\alpha - \cos\epsilon \sin\lambda\theta \sin\alpha \sin\delta')$$

$$\frac{d(\delta' - \delta)}{d\lambda\theta} = 0 \Rightarrow \pi (\sin\epsilon \cos\lambda\theta \cos\delta' + \sin\lambda\theta \sin\delta' \cos\alpha - \cos\epsilon \cos\lambda\theta \sin\alpha \sin\delta') = 0$$

$$\lambda\theta = 0$$

$$\Rightarrow \sin\epsilon \cos\delta' = \cos\epsilon \sin\alpha \sin\delta'$$

$$\tan\epsilon = \sin\alpha \tan\delta'$$

$$\sin\delta \sec\delta' = \sin\epsilon \sin\lambda\theta$$

$$\cos\delta \sec\delta' \cos\alpha \cos\epsilon = \cos\lambda\theta$$

$$\cos\delta \sec\delta' \sin\alpha \sin\epsilon = \cos\delta \cot\epsilon \cot\delta \sec\delta' = \sin\delta \cot\epsilon$$

$$= \sin\epsilon \sin\lambda\theta \cot\epsilon = \cos\epsilon \sin\lambda\theta$$

Nakon izraz kojim se može odrediti $\lambda\theta$ u trenutku kada je uticaj GP na α zvezde maksimalan. Koliko je $(\alpha' - \alpha)_{\max}$?

$$\alpha' - \alpha = \pi \sin(\alpha - \alpha_0) \cos\delta \sec\delta'$$

$$\tan\lambda\theta = -\cot\alpha \cos\epsilon$$

$$\sin\lambda\theta = \frac{\sin\lambda\theta}{\cos\lambda\theta} = \frac{\tan\lambda\theta}{\sqrt{1 + \tan^2\lambda\theta}}$$

$$1 + \tan^2\lambda\theta = 1 + \frac{\sin^2\lambda\theta}{\cos^2\lambda\theta} = \frac{1}{\cos^2\lambda\theta}$$

$$\cos\lambda\theta = \left(\frac{1}{\cos\lambda\theta}\right)^{-1} = \frac{1}{\sqrt{1 + \tan^2\lambda\theta}}$$

od range:

$$\alpha' - \alpha = -\pi \sec\delta' (\sin\alpha \cos\lambda\theta - \cos\alpha \cos\epsilon \sin\lambda\theta)$$

$$(\alpha' - \alpha)_{\max} = -\pi \sec\delta' \frac{1}{\sqrt{1 + \tan^2\lambda\theta}} (\sin\alpha - \cos\alpha \cos\epsilon \tan\lambda\theta)$$

$$(\alpha' - \alpha)_{\max} = -\pi \sec\delta' \frac{1}{\sqrt{1 + \cot^2\alpha \cos^2\epsilon}} (\sin\alpha + \cos\alpha \cos\epsilon \frac{\cos\alpha}{\sin\alpha} \cos\epsilon)$$

$$(\alpha' - \alpha)_{\max} = -\pi \sec\delta' \frac{\sin\alpha + \cos^2\alpha \cos^2\epsilon \frac{1}{\sin\alpha}}{\sqrt{\sin^2\alpha + \cos^2\alpha \cos^2\epsilon}} = -\pi \sec\delta' \frac{\sin^2\alpha + \cos^2\alpha \cos^2\epsilon}{(\sin^2\alpha + \cos^2\alpha \cos^2\epsilon)^{3/2}}$$

$$(\alpha' - \alpha)_{\max} = -\pi \sec\delta' \frac{1}{\sqrt{1 + \frac{\cos^2\epsilon}{\sin^2\alpha} \cos^2\epsilon}} (\dots)$$

$$= -\pi \sec\delta' \frac{\sin\alpha}{\sqrt{\sin^2\alpha + \cos^2\alpha \cos^2\epsilon}} (\dots)$$

$$\Rightarrow (\alpha' - \alpha)_{\max} = -\pi \sec\delta' \sqrt{1 - \cos^2\alpha \sin^2\epsilon}$$

$$\sin^2\alpha + \cos^2\alpha \cos^2\epsilon = 1 - \cos^2\alpha + \cos^2\alpha \cos^2\epsilon = 1 - \cos^2\alpha (1 - \cos^2\epsilon) = 1 - \cos^2\alpha \sin^2\epsilon$$

$$\Rightarrow (\alpha' - \alpha)_{\max} = -\pi \sec\delta' \sqrt{1 - \cos^2\alpha \sin^2\epsilon}$$

Odredi ložišdu Sunca u trenutku kada je u blizini CP na d maksimalu

$$\text{tg } \ell = \sin \alpha \text{tg } \delta$$

$$\delta' - \delta = \pi [\sin \delta \cos \delta - \cos \delta \sin \delta \cos (\alpha_0 - \alpha)]$$

$$\delta' - \delta = \pi [\sin \epsilon \sin \lambda_0 \cos \delta - \sin \delta \cos \alpha \cos \lambda_0 + \cos \epsilon \sin \lambda_0 \sin \delta \sin \alpha]$$

$$\frac{d(\delta' - \delta)}{d\lambda_0} = 0 \Rightarrow \sin \epsilon \cos \lambda_0 \cos \delta + \sin \delta \cos \alpha \sin \lambda_0 - \cos \epsilon \cos \lambda_0 \sin \delta \sin \alpha = 0$$

$$\Rightarrow \sin \epsilon \cos \delta = \cos \epsilon \sin \delta \sin \alpha$$

Ali ovde: $\sin \epsilon \cos \lambda_0 \cos \delta + \sin \delta \cos \alpha \sin \lambda_0 - \cos \epsilon \cos \lambda_0 \sin \delta \sin \alpha = 0$
 $\sin \lambda_0 \sin \delta \cos \alpha = \cos \lambda_0 (\cos \epsilon \sin \delta \sin \alpha - \sin \epsilon \cos \delta)$

$$\text{tg } \lambda_0 = \frac{\cos \epsilon \sin \delta \sin \alpha - \sin \epsilon \cos \delta}{\cos \alpha \sin \delta}$$

$$\text{tg } \lambda_0 = \cos \epsilon \text{tg } \alpha - \sin \epsilon \text{ctg } \delta \sin \alpha$$



$$P = \sin T \cdot \pi$$

$$\frac{a}{r} = \text{tg } P \approx P$$

ako je $P = P'' \rightarrow r = 1 \text{ pc}$ - šta je 1 pc?

$$a = 149.6 \cdot 10^6 \text{ km} = 1 \text{ AU}$$

$$1 \text{ pc} = \frac{a}{\sin P''} = d$$

$$\pi [\text{rad}] = \frac{1}{r [\text{AU}]}$$

$$P = \frac{a}{r \sin P''} \sin T$$

$$\pi'' = \frac{a}{r \sin P''} = \frac{d}{r} = \frac{1}{r/d}$$

$$\pi'' = \frac{1}{r/d} \rightarrow r [\text{pc}] = \frac{1}{\pi''}$$

Koliko je od nas udaljena zvezda $\alpha \text{ Cen}$ ($\pi = 0''.75$). Rezultat izraziti u pc, m

$$\pi'' = \frac{1}{r}$$

$$1 \text{ pc} = 206265 \text{ AU}$$

$$\sin P'' \approx \frac{1}{206265}$$

$$r [\text{pc}] = \frac{1}{\pi''} = 1.333 \text{ pc}$$

$$r = 4.113 \cdot 10^{16} \text{ m}$$

$$1 \text{ pc} = 3.26 \text{ ly}$$

$$1 \text{ pc} = 3.086 \cdot 10^{13} \text{ km}$$

Uporediti maksimalna rastojanja koja mogu da

bude određenim satelitskim Hiparhos i Gaia misijama

$$\pi_H = 0''.00002$$

$$\pi_H = 0''.002$$

$$r_H [\text{pc}] = 500 \text{ pc}$$

$$r_G [\text{pc}] = 50000 \text{ pc}$$

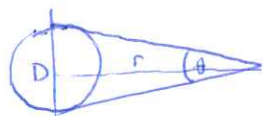
• Određivanje veličina nebeskih tela

P. Suncu $\theta_0 = 32''$

$$r = 149.6 \cdot 10^6 \text{ km}$$

$$D = 1.393 \cdot 10^6 \text{ km}$$

$$D/2 = R \approx 696 \text{ 000 km}$$

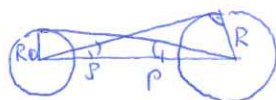


θ - ugledni prečnik objekta

r - rastojanje do objekta

D - prečnik

$$D = \frac{\theta ['] r}{206265}$$



$$\sin p = \frac{R \theta}{D} \rightarrow R \theta = D \sin p$$

$$\sin \beta = \frac{R}{D} \rightarrow R = D \sin \beta$$

$$\Rightarrow \frac{R}{R \theta} = \frac{\sin \beta}{\sin p} \approx \frac{\beta}{p}$$

$$R = R \oplus \frac{p}{\beta}$$

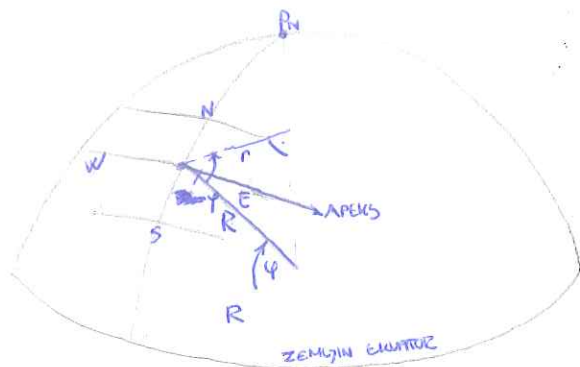
$$R \oplus = 6378 \text{ km}$$

$$\text{Merkur: } \beta = 15' 32''.6$$

$$p = 57' 2''.7$$

$$R = 1734.87 \text{ km}$$

Podpalcia Zemljine rotacije oko svoje os jeste dnevna aberacija



$$\sin(90^\circ - \varphi) = \frac{r}{R}$$

$$r = R \cos \varphi$$

$$\omega \varphi = \frac{2\pi R}{P}$$

$$\omega = \frac{2\pi R}{P}$$

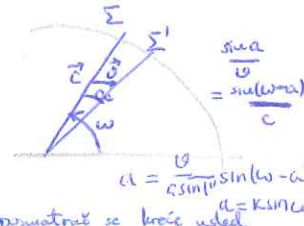
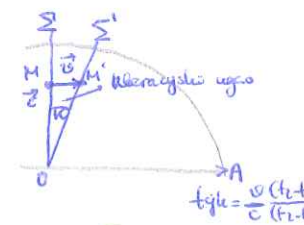
$$\omega \varphi = \frac{2\pi R \cos \varphi}{P} = v \cos \varphi$$

$$\omega = 460 \frac{\text{m}}{\text{s}} \approx 1.6 \cdot 10^6 \frac{\text{m}}{\text{s}}$$

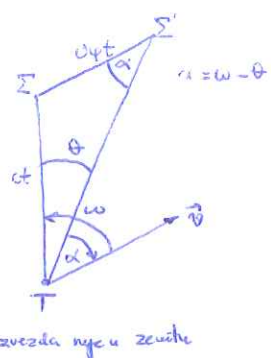
ω - period rotacije Zemlje

P - brzina na ekvatoru (tangencijalna)

Brzina, γ Druvina $\Delta \delta = \pm 20''$



Aberacija kao pojav nastaje jer je brzina svetlosti konstantna (prolazi kroz tubus teleskopa za neko konstantno vreme) - posmatrač se kreće usled rotacije i revolucije



$\Delta \Sigma \Sigma' T$

$$\frac{\sin \theta}{\omega \varphi t} = \frac{\sin(\omega - \theta)}{ct}$$

$$\theta \approx \sin \theta = \frac{\omega \varphi}{c} \sin(\omega - \theta)$$

$$\theta \approx \frac{\omega \varphi}{c} \sin \omega$$

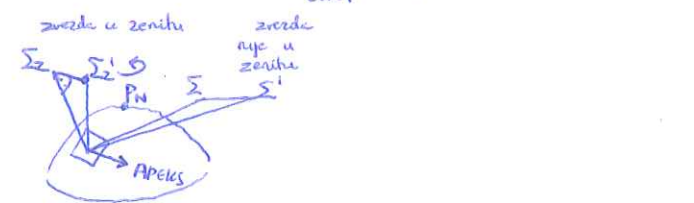
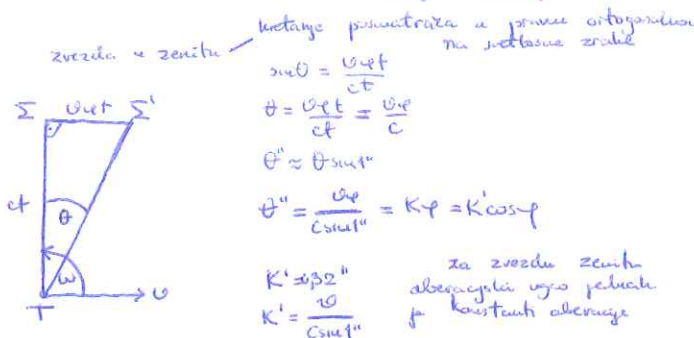
$$\theta'' = \theta \sin 1''$$

$$\theta'' = \frac{\omega \varphi}{c \sin 1''} \sin \omega$$

$K \varphi$ - konstanta aberacije

$$K \varphi = \frac{\omega \varphi}{c \sin 1''} = \frac{\omega \cos \varphi}{c \sin 1''} = K' \cos \varphi$$

konstanta dnevne aberacije na ekvatoru



Dakle, usled dnevne aberacije, položaj zvezde biće pomeren ka istočnoj tački za mali ugao

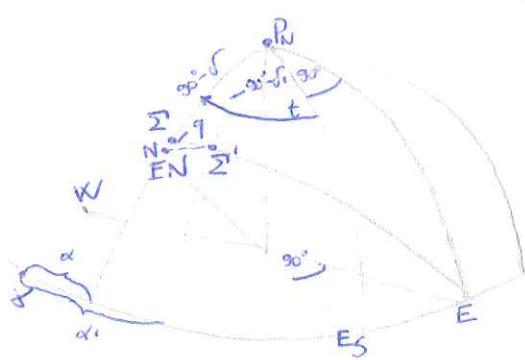
$$\Delta \Sigma \Sigma' = 32'' \cdot \cos \varphi \cdot \sin \omega$$

Uticaj postoji na ekvatorske koordinate!

Izvesti uticaj dnevne aberacije na ekvatorske koordinate.

$\Sigma E = \omega$

$\Delta \Sigma \Sigma' N$



$$\left(\delta' - \delta \right) \cos \delta = \Sigma N = \Sigma \Sigma' \cos \varphi = K \varphi \sin \omega \cos \varphi$$

$$\left(\alpha' - \alpha \right) \cos \delta = \Sigma' N = \Sigma \Sigma' \sin \varphi = K \varphi \sin \omega \sin \varphi$$

$$\frac{\sin \omega}{\sin(180^\circ - \varphi)} = \frac{\sin 90^\circ}{\sin(90^\circ - \delta)} = \frac{1}{\sin \delta}$$

$$\sin \omega \sin \varphi = \cos \delta \quad (1)$$

$$\sin \omega \cos(180^\circ - \varphi) = \cos 90^\circ \sin(90^\circ - \delta) - \sin 90^\circ \cos(90^\circ - \delta) \cos(90^\circ + t)$$

$$-\sin \omega \cos \varphi = \sin \delta \sin t$$

$$\sin \omega \cos \varphi = -\sin \delta \sin t \quad (2)$$

$$(1) : (2) \rightarrow (*) \Rightarrow$$

$$\delta' - \delta = -K \varphi \sin \delta \sin t$$

$$\alpha' - \alpha = K \varphi \cos \delta \cos t$$

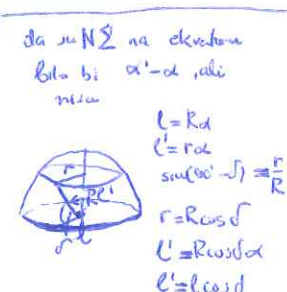
$$\delta' - \delta = K \varphi \sin \delta \sin t$$

$\Delta \alpha$ maksimalan u meridijanu, $\Delta \delta = 0$ (obrnuto za $t = 6^h$)

Meridijanski prolaz kroz meridijan se moglo odobavljati uticaja DA (ovo se vazi za Σ doljine)

nije ni potreban:

$\Delta N \Sigma' P N$



da su $N \Sigma$ na ekvatoru bilo bi $\alpha' - \alpha$, ali nista

$$l = R \delta$$

$$l' = R \alpha$$

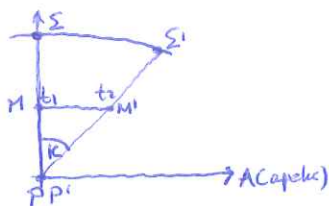
$$\sin(90^\circ - \delta) = \frac{r}{R}$$

$$r = R \cos \delta$$

$$l' = R \cos \delta \alpha$$

$$l' = l \cos \delta$$

Aberacija

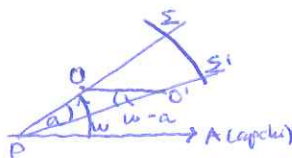


K - konstantna odsvajanje

$$\tan K = \frac{MM'}{MP'} = \frac{v(t_2 - t_1)}{c(t_2 - t_1)} = \frac{v}{c}$$

$$\frac{\sin K}{\cos K} = \frac{v}{c}$$

$$K \sin K' = \frac{v}{c} \Rightarrow K = \frac{v}{c \sin K'}$$



$$\frac{\sin a}{\sin a'} = \frac{\sin(\omega - a)}{\sin \omega}$$

$$\sin a = \sin(\omega - a) \frac{\sin \omega}{\sin \omega'} = \sin(\omega - a) \frac{v(t_2 - t_1)}{c(t_2 - t_1)} = \sin(\omega - a) \frac{v}{c}$$

$$a = \frac{v}{c \sin \omega'} \sin(\omega - a) \approx \frac{v}{c \sin \omega'} \sin \omega$$

$$a = K \sin \omega$$

dužina: $K = \frac{v_e}{c \sin \omega'}$, $v_e = 460 \text{ m/s}$

$$v_e = \frac{2\pi R \Theta}{T}$$

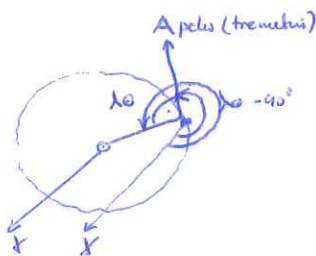
$$\omega \varphi = v_e \cos \varphi$$

$$K \varphi = K \cos \varphi$$

$$K = 0.32$$

$$\Sigma' = K \sin \omega = K \cos \varphi \sin \omega$$

godišnja $K = 20.5$



$$\omega \approx \frac{2\pi a}{T} = 30 \text{ km/s}$$

$$K = \frac{v}{c \sin \omega'} = 20.52$$

$$\lambda_A = \lambda_0 - 90^\circ$$

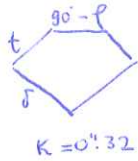
Koliko je uticaj dnevne oborine na α, δ zvezde $\Sigma (\alpha = 9^h 48^m 13^s.2, \delta = 63^\circ 36' 15''.8)$ u trenutku NID za posmatralca u Beogradu ($\varphi = 44^\circ 48' 13''.2$)

$$\varphi = 44.8036667$$

$$\delta = 63.604388$$

$$\Delta\alpha = k \cos \varphi \cos \delta \sec \delta$$

$$\Delta\delta = k \sin \varphi \sin \delta \sec \delta$$



$$\cos t = \cos(90^\circ - \delta) \cos \alpha$$

$$\cos t = \frac{\sin \delta}{\sin \alpha} \rightarrow t = 60.467373 = 60^\circ 28' 2''.54$$

$$\Delta\alpha = 0.0168$$

$$\Delta\delta = 0.1769$$

Uticaj DA na α, δ $\Sigma (\alpha = 18^h 36^m 34^s.09, \delta = 35^\circ 46' 24''.52)$ u trenutku SEV = 22h, u Beogradu ($\varphi = 44^\circ 48' 13''.2, \lambda = 20^\circ 27' 44''.1$); $S_0 = 10^h 58^m 55^s.96$

$$SEV = t + UT$$

$$s = s_0 + t_s (1 + p)$$

$$\alpha + t = s_0 - \lambda p + (UT + \lambda) (1 + p)$$

$$t = s_0 - \lambda p + (SEV - t + \lambda) (1 + p) - \alpha$$

$$p = \frac{1}{365.2422}$$

$$\Delta\alpha, \Delta\delta$$

$$\Delta\alpha = -0.017$$

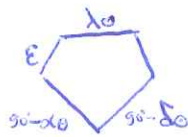
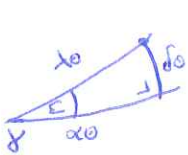
$$\Delta\delta = 0.157$$

1sr. dan $\approx (1 + p) \approx$ v. dan

Koliko je uticaj GP na α zvezde u trenutku kada je uticaj GA na α maksimalan?

$$GP: \alpha' - \alpha = \pi \sin(\alpha - \alpha_0) \cos \delta \sec \delta$$

$$= \pi \sec \delta (\sin \alpha \cos \delta \cos \delta - \cos \alpha \sin \alpha \cos \delta)$$



$$\cos \lambda_0 = \cos \alpha \cos \delta$$

$$\sin \lambda_0 \cos \delta = \cos \epsilon \sin \lambda_0$$

$$\sin \delta = \sin \epsilon \sin \lambda_0 / \cos \epsilon$$

$$\frac{\cos \epsilon \sin \delta}{\sin \epsilon} = \cos \epsilon \sin \lambda_0$$

$$\cos \epsilon \sin \delta = \cos \epsilon \sin \lambda_0$$

$$\cos \epsilon \cdot \cos(90^\circ - \delta) = \sin \alpha \cos \delta$$

$$\cos \epsilon \cdot \cos \delta = \sin \alpha \cos \delta$$

$$\cos \epsilon \sin \delta = \sin \alpha \cos \delta$$

$$\sin \alpha \cos \delta = \cos \epsilon \sin \lambda_0$$

$$\Rightarrow \alpha' - \alpha = \pi \sec \delta (\sin \alpha \cos \lambda_0 - \cos \alpha \cos \epsilon \sin \lambda_0) \quad (1)$$

$$GA: \frac{d\alpha}{dt} = 0 \rightarrow \cos \lambda_0 \sin \alpha - \sin \lambda_0 \cos \alpha \cos \epsilon = 0 \quad (2)$$

$$\alpha' - \alpha = -k \sec \delta (\sin \lambda_0 \sin \alpha + \cos \lambda_0 \cos \alpha \cos \epsilon)$$

$$(1) \text{ i } (2) \rightarrow (\alpha' - \alpha)_{GP} = 0$$

Ali je uticaj GA u nekom trenutku na α najveći, pokazati da je u tom trenutku zvezda na deklinaciji. Suma α_A i deklinacija zvezde

$$\alpha' - \alpha = -k \sec \delta (\sin \lambda_0 \sin \alpha + \cos \lambda_0 \cos \alpha \cos \epsilon)$$

$$\frac{d(\alpha' - \alpha)}{dt} = 0 \rightarrow \cos \lambda_0 \sin \alpha + \cos \lambda_0 \cos \alpha \cos \epsilon = 0$$

$$\cos \lambda_0 \cos \alpha = -\cos \epsilon \sin \alpha$$

$$\cos \lambda_0 \cos \alpha = -\cos \epsilon \sin \alpha \quad (1) \rightarrow \cos \epsilon = \frac{\cos \lambda_0 \cos \alpha}{\sin \alpha} = \cos \lambda_0 \cot \alpha$$

$$(1) \text{ i } (2) \Rightarrow \cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

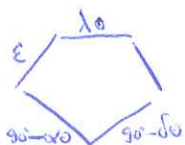
$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

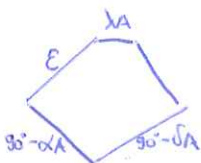


$$\cos \epsilon = \cos \lambda_0 \cot \alpha \quad (2)$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\Rightarrow \cos \epsilon = \cos \lambda_0 \cot \alpha$$



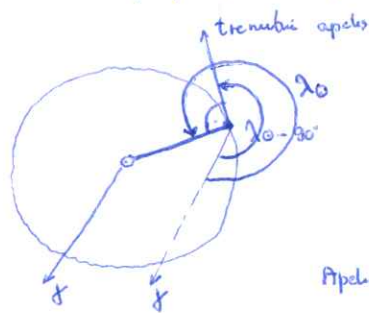
$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha \quad (3)$$

$$\cos \epsilon = \cos \lambda_0 \cot \alpha$$

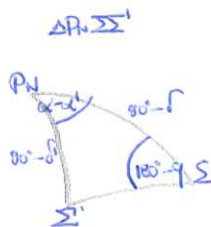
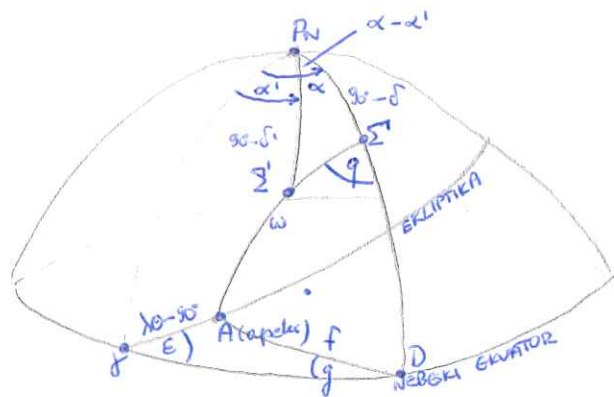
Izvesti utjecaj godišnje aberacije na ekliptičke koordinate.



GA je dešavanje zbog Zemljinog kretanja oko Sunca
 $\frac{v}{c} \approx \frac{2\pi a}{T} \approx 30 \text{ km/s}$ - brzina Zemljinog kretanja
 - velika polusena
 - siderična godina

$$K = \frac{v}{c \sin i} \approx 20''.5, \lambda_A = \lambda_0 - 90^\circ$$

Apels - tačka na ekliptici čija je ekliptička longitududa male udaljenosti za 90° od λ_0



$$\frac{\sin(\alpha - \alpha')}{\sin \Sigma \Sigma'} = \frac{\sin(180^\circ - q)}{\sin(90^\circ - \delta')}$$

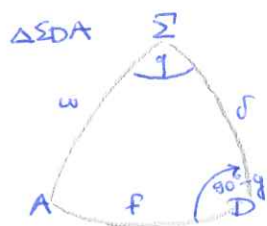
$$\frac{\sin(\alpha - \alpha')}{\sin \Sigma \Sigma'} = \frac{\sin q}{\cos \delta'}$$

$$\sin(\alpha - \alpha') \approx \alpha - \alpha'$$

$$\cos \delta' \approx \cos \delta$$

$$\Sigma \Sigma' = K \sin w$$

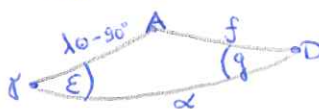
$$\Rightarrow (\alpha - \alpha') \cos \delta = K \sin w \sin q$$



$$\frac{\sin w}{\sin(90^\circ - q)} = \frac{\sin f}{\sin q}$$

$$\sin w \sin q = \cos \gamma \sin f$$

ΔADγ



$$\sin f \cos \gamma = \cos(\lambda_0 - 90^\circ) \sin \alpha - \sin(\lambda_0 - 90^\circ) \cos \alpha \cos \epsilon$$

$$\sin f \cos \gamma = \sin \lambda_0 \sin \alpha + \cos \lambda_0 \cos \alpha \cos \epsilon$$

$$\Rightarrow (\alpha' - \alpha) = \sec \delta K \sin w \sin q = K \sec \delta \cos \gamma \sin f = K \sec \delta (\sin \lambda_0 \sin \alpha + \cos \lambda_0 \cos \alpha \cos \epsilon)$$

$$\alpha' - \alpha = K \sec \delta (\sin \lambda_0 \sin \alpha + \cos \lambda_0 \cos \alpha \cos \epsilon)$$

ΔPΣΣ'

$$\sin \Sigma \Sigma' \cos(180^\circ - q) = \cos(90^\circ - \delta') \sin(90^\circ - \delta) - \sin(90^\circ - \delta') \cos(90^\circ - \delta) \cos(\alpha - \alpha')$$

$$\sin \Sigma \Sigma' \approx \Sigma \Sigma' = K \sin w$$

$$\cos(180^\circ - q) = -\cos q$$

$$\cos(\alpha - \alpha') \approx 1$$

$$\Rightarrow -K \sin w \cos q = -\sin(\delta' - \delta) = \sin(\delta' - \delta) \approx \delta' - \delta$$

$$\Rightarrow \delta' - \delta = -K \sin w \cos q$$

ΔΣDA

$$\sin w \cos q = \cos f \sin \delta - \sin f \cos \delta \cos(90^\circ - q)$$

$$\sin w \cos q = \cos f \sin \delta - \sin f \cos \delta \sin q$$

ΔADγ

$$\cos f = \cos \alpha \cos(\lambda_0 - 90^\circ) + \sin \alpha \sin(\lambda_0 - 90^\circ) \cos \epsilon$$

$$\cos f = \cos \alpha \sin \lambda_0 - \sin \alpha \cos \lambda_0 \cos \epsilon$$

$$\cos(\lambda_0 - 90^\circ) = \sin \lambda_0, \sin(\lambda_0 - 90^\circ) = -\cos \lambda_0$$

$$\frac{\sin f}{\sin \epsilon} = \frac{\sin(\lambda_0 - 90^\circ)}{\sin q} = -\frac{\cos \lambda_0}{\sin q}$$

$$\sin f \sin q = -\cos \lambda_0 \sin \epsilon$$

$$\delta' - \delta = -K (\cos f \sin \delta - \sin f \cos \delta \sin q)$$

$$= -K [\sin \delta (\cos \alpha \sin \lambda_0 - \sin \alpha \cos \lambda_0 \cos \epsilon) - \cos \delta (-\cos \lambda_0 \sin \epsilon)]$$

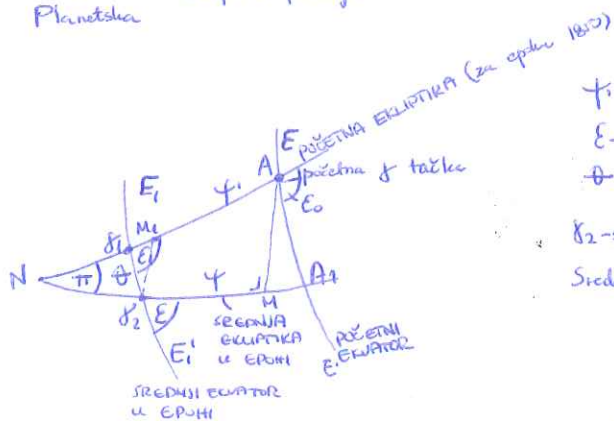
$$= -K (\sin \delta \cos \alpha \sin \lambda_0 - \sin \alpha \cos \lambda_0 \cos \epsilon \sin \delta + \cos \delta \cos \lambda_0 \sin \epsilon)$$

$$\delta' - \delta = -K [\sin \lambda_0 \cos \alpha \sin \delta + \cos \lambda_0 \cos \epsilon (\tan \delta \cos \delta - \sin \alpha \sin \delta)]$$

Hipoteza jeotermične da toplote gnojine traje kratko od silikata, što je posledica precesije (retrogradno pomeranje γ tačke po eliptici)

Luni-solarna
+
Planetska = Opšte precesije

Severnojke $\rightarrow$ Vega



γ - luni-solarna precesija ($\gamma \rightarrow \delta_1$)

ϵ - srednji nagib eliptike u epohi

θ - planetska precesija ($\delta_1 \rightarrow \delta_2$)

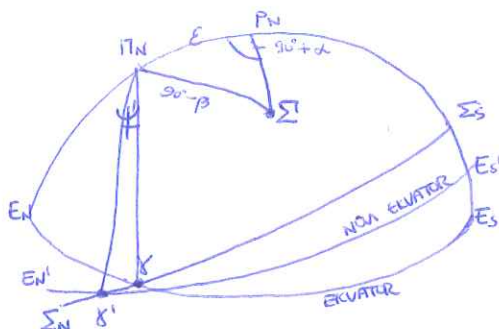
δ_2 - srednja γ tačka

Srednja eliptika, dvatale i γ tački odgovaraju srednji polari i srednji položaj nebeskih tela na nebeskoj sferi

$N\gamma = NM$ - longituda uzduž žirna N srednje eliptike na položaj eliptike

$N\gamma_2 = NM_1$

LS oko $50'.37$ godišnje
planetska oko 0.13 u zadržavanju suseda



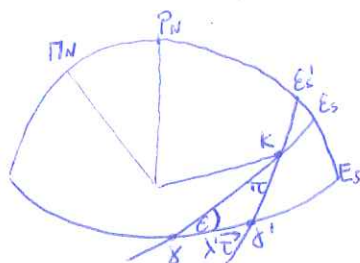
$T = 1 \text{ god}$
 $\psi = 50'.37$

$T = \frac{360'}{50'.37} \approx 25100 \text{ god}$
planetska
godine

Uticaj LS precesije na eliptičke koordinate
 $dh = \psi T$
 $dp = 0$

Opšte:
 $\gamma = \psi - \theta \cos \epsilon_1$ nagib eliptike u odnosu na pravi elip
luni solarna
planetska

Planetska precesija - 60000 god period $\rightarrow$ pomeru ϵ , δ nagiba eliptike prema dvatale



$\pi = 0'.47 / \text{god}$ $\leftarrow$ Nagib spinde
 $\lambda' = 0''.17 / \text{god}$ $\leftarrow$ brzoje gajur tačke
 $d\alpha = -\lambda' T$
 $d\delta = 0$

Precesioni kraj spindla su srednje i dupljenjem od $6 \cdot 10^4 \text{ god}$

Opšte precesije $p = 50'.25$

$$\sin \delta = \cos \epsilon \sin \beta + \sin \epsilon \cos \beta \sin \lambda \quad (1)$$

$$\cos \beta \cos \lambda = \cos \delta \cos \alpha \quad (2)$$

$$\begin{aligned} \cos \theta d\theta &= \sin \theta \cos \phi \cos \lambda d\lambda \\ \cos \theta d\theta &= \sin \theta \cos \phi \cos \alpha d\alpha \\ d\theta &= \sin \theta \cos \alpha d\alpha \\ d\lambda &= 4\pi \Rightarrow \boxed{d\Omega = 4\pi} \end{aligned}$$
$$-\sin \alpha \cos \alpha d\alpha - \cos \alpha \sin \alpha d\alpha = -\cos \alpha \sin \alpha d\alpha$$

$$\cos \beta \sin \alpha = \sin \beta \sin \alpha \epsilon + \cos \beta \cos \alpha \sin \alpha d$$

$$\rightarrow \sin \theta \cos \alpha d\theta + \cos \theta \sin \alpha d\alpha = (\sin \theta \sin \epsilon + \cos \theta \cos \epsilon \sin \alpha) d\lambda$$

$$\cos \delta \sin \alpha d\alpha = (\sin \delta \sin \epsilon + \cos \delta \cos \epsilon \sin \alpha) \psi_T - \sin \delta \cos^2 \alpha \sin \epsilon \psi_T$$

$$\cos\theta \sin\alpha \, d\alpha = \frac{4\pi}{3} (\cancel{\sin\theta \sin\alpha} + \cos\theta \cos\alpha \sin\alpha - \cancel{\sin\theta \sin\alpha} + \sin\theta \sin\alpha \cos\alpha)$$

$$\Rightarrow |d\alpha| = 4\pi (\cos \epsilon + \tan \epsilon \sin \epsilon \sin \alpha)$$

$$d\delta = 4\tau \cos\beta$$

$$d\phi = \psi \tau \sin \theta \cos \theta d\theta$$

$$\cos \beta = \sin E \sin (90^\circ + \alpha) = \sin E \cos \alpha$$

$$d\delta = \psi_T \cos \beta$$

$$\left. \begin{aligned} d\alpha &= \psi \tau (\cos \epsilon + \tan \epsilon \sin \epsilon \sin \alpha) \\ \cos(90^\circ + \alpha) &= \cot \epsilon \cot \epsilon = -\sin \alpha \end{aligned} \right\} \Rightarrow \begin{aligned} d\alpha &= \psi \tau (\cos \epsilon - \tan \epsilon \sin \epsilon \cot \epsilon \cot \epsilon) \\ d\alpha &= \psi \tau (\cos \epsilon - \cos \epsilon) = 0 \end{aligned}$$

$$d\alpha = 4\pi(\omega_S E - \omega_S E) = 0$$

(4)

$$\cos^2 \int \sec \alpha \, d\alpha = \sec \int d\alpha = \sec \lambda \, d\lambda$$

$$\sin \alpha = dy \cdot \tan \delta / d''$$

$$\cos \alpha d\alpha = \frac{\csc \alpha}{\sin \alpha \csc \alpha} \frac{1}{\cos^2 \delta} d\delta$$

$$\Rightarrow \cos \alpha d\alpha = \sin \alpha \csc \theta \delta \frac{1}{\cos^2 \theta} d\theta$$

$$\frac{dy}{dx} \propto \frac{dx}{dx} \cos^2 \delta = \frac{dy}{d\delta} d\delta$$

$$\sin \delta = \sin \epsilon \sin \lambda / d''$$

$$\cos \delta d\delta = \frac{\sin \epsilon \cos \lambda d\lambda}{\sin \delta / \sin \lambda}$$

$$\cos \delta d\delta = \frac{\sin \delta}{\sin \lambda} \cos \lambda d\lambda$$

$$\Rightarrow \text{ctg} \delta d\delta = \text{ctg} \lambda d\lambda$$

$$(2) \quad \begin{aligned} d\alpha &= 4\pi (\cos E + \tan^2 E \sin E) \\ d\delta &= 4\pi \cos \alpha \sin E \end{aligned}$$

$$d\alpha = 4\pi \cos \epsilon + 4\pi b y \int \sin \epsilon \sin \alpha \, / - \cos \alpha$$

$$\cos \alpha d\alpha = 4\pi \cos \alpha \cos E + \underbrace{4\pi \cos \alpha \sin E}_{dE} \log \sin \alpha$$

$$\sin \alpha = \csc \alpha \text{ \& } \tan \alpha$$

$$\sin \alpha \sin \epsilon \cot \gamma = \cos \epsilon$$

$$\Rightarrow \cos \alpha d\alpha = \frac{1}{4\pi \sin \epsilon} \cos \alpha \sin \alpha d\epsilon + \frac{1}{4\pi} \sin \alpha d\epsilon$$

$$ct_y d\alpha = (ty^{\sigma} + ct_y^{\sigma}) d\sigma = \frac{1}{\cos\delta \sin\delta} d\sigma / \cos^2\delta$$

$$\Rightarrow \underline{c \tan \alpha \cos^2 \delta \, d\alpha = c \tan \delta \, d\delta}$$

$$d\delta = 4\pi \cos \alpha \sin \epsilon = d\lambda \cos \alpha \sin \epsilon$$

$$\cos \lambda = \cos \alpha \cos \delta$$

$$\cos \alpha = \frac{\cos \lambda}{\cos \phi}$$

$$\cancel{\cos \delta} d\delta = d\lambda \sin \epsilon \frac{\cos \lambda}{\cos \delta}$$

$$\cos \theta d\theta = d\lambda \sin \theta \cos \lambda$$

$$\underline{\operatorname{ctg} \delta d\delta = \operatorname{ctg} \lambda d\lambda}$$

$$\frac{\sin \delta}{\sin \lambda} = \sin \epsilon$$

Náci formule pomocí kojích se mají izračunati gubšije pravižne pravogilne koordinatu ušed opšte preeže. Mide se na pravogilne elvatorilne koordinatu

$$d\alpha = \tau(m + n \operatorname{tg} \delta \sin \alpha)$$

$$d\delta = \tau \cos \alpha$$

$$x = \cos \delta \cos \alpha$$

$$y = \cos \delta \sin \alpha$$

$$z = \sin \delta$$

$$\frac{dz}{dt} = \cos \delta \frac{d\delta}{dt} = \tau \cos \alpha \cos \delta = \tau n x$$

$$\frac{dy}{dt} = -\sin \delta \sin \alpha \frac{d\delta}{dt} + \cos \delta \cos \alpha \frac{d\alpha}{dt}$$

$$\frac{dy}{dt} = -\tau n \sin \delta \sin \alpha \cos \alpha + \cos \delta \cos \alpha \tau m + \tau n \cos \delta \cos \alpha \operatorname{tg} \delta \sin \alpha$$

$$\frac{dy}{dt} = -\tau n \sin \delta \sin \alpha \cos \alpha + \tau m \cos \delta \cos \alpha + \tau n \sin \delta \cos \alpha \sin \alpha$$

$$\Rightarrow \frac{dy}{dt} = \tau m \cos \delta \cos \alpha = \tau m x$$

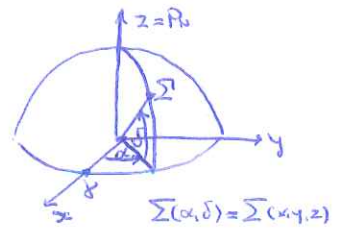
$$\frac{dx}{dt} = -\sin \delta \cos \alpha \frac{d\delta}{dt} - \cos \delta \sin \alpha \frac{d\alpha}{dt}$$

$$\frac{dx}{dt} = -\sin \delta \cos \alpha \tau n - \tau m \cos \delta \sin \alpha - \tau n \cos \delta \sin \alpha \operatorname{tg} \delta \sin \alpha$$

$$\frac{dx}{dt} = -\sin \delta \cos \alpha \tau n - \tau m \cos \delta \sin \alpha - \tau n \sin \delta \sin \alpha \cos \alpha$$

$$\frac{dx}{dt} = -\tau m y - \sin \delta \tau n$$

$$\frac{dx}{dt} = -\tau m y - \tau n z$$



$$dx_i = \Psi' \tau (\cos \ell + \operatorname{tg} \delta \sin \ell \sin \alpha)$$

$$d\alpha = -\lambda' \tau$$

$$d\alpha = \tau \left(\underbrace{\Psi' \cos \ell - \lambda'}_m + \Psi' \operatorname{tg} \delta \sin \ell \sin \alpha \right)$$

$$\Psi' \sin \ell = n$$

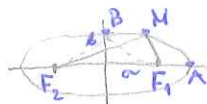
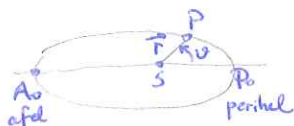
$$d\alpha = \tau (m + n \operatorname{tg} \delta \sin \alpha)$$

$$d\delta = \tau \cos \alpha$$

Keplerovi zakoni

I: svaka planeta se kreće po eliptičnoj putanji, t.d. x u jednoj od žiža Sunce

$(\vec{r}, \nu)$ - planarne koordinate



$$E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$F_1(0, c)$$

$$F_2(0, -c)$$

$$F_1M + F_2M = 2a$$

$$F_1B + F_2B = 2a$$

$$F_1B = \sqrt{c^2 + b^2}$$

$$F_2B = \sqrt{c^2 + b^2}$$

$$\Rightarrow 2\sqrt{c^2 + b^2} = 2a$$

$$a^2 = b^2 + c^2$$

$$F_1A + F_2A = \text{const.}$$

$$F_1A + F_2A = (a+c) + (a-c) = 2a$$

$$\text{ekscentricitet: } e = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a} = \sqrt{1 - \frac{b^2}{a^2}}$$

$$0 \leq e \leq 1$$

II: radius vektor Sunce - planeta opisuje u jednakim vremenskim intervalima jednake površine

$$\frac{a^3}{T^2} = \text{const.}$$

$$a[A], T[y] \rightarrow a^3 = T^2$$

$$\text{Newtonova generalizacija: } \frac{a^3}{T^2} = \frac{G}{4\pi^2} (m_1 + m_2)$$

$$G = 6.67 \cdot 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

$$F = G \frac{m_1 m_2}{r^2}$$

Ograničene (eliptične) putanje

r_p - rastojanje u perihelu

r_a - rastojanje u afelu

$$a = \frac{r_p + r_a}{2}$$

$$r_p = a(1-e)$$

$$r_a = a(1+e)$$

Nekle konstante

$$M_{\odot} = 1.989 \cdot 10^{30} \text{ kg}$$

$$M_{\oplus} = 5.97 \cdot 10^{24} \text{ kg}$$

$$T_{\oplus} = 365.2564 \text{ d}$$

$$a_{\oplus} = 149.6 \cdot 10^6 \text{ km}$$

$$r_{\oplus} = 6378 \text{ km}$$

ukupna površina elipse je πab , radius vektor se pobraže za orb. period T , a ova je jednaka dvostrukoj sektorskoj brzini; $h = 2\pi ab/T$

$$T = \frac{2\pi ab}{h} = \frac{2\pi a \cdot a\sqrt{1-e^2}}{h} = 2\pi \sqrt{\frac{a^3}{Gm}} \quad (m = m_1 + m_2)$$

$$e^2 = 1 - \frac{b^2}{a^2}$$

$$b^2 = a^2(1-e^2)$$

orbitalni period se zavisi od e

$$n = \frac{2\pi}{T} - \text{srednja dnevna brzina (prosječna ugaona brzina tela na orbiti)}$$

$$T = 2\pi \sqrt{\frac{a^3}{Gm}}$$

$$n [^\circ/\text{dan}], n = \frac{360}{T}$$

pr. Jupiter obilazi oko Sunca za 11.86 godina, tj za 4331.87 sr. dana

$$n_{\text{Jup}} \approx 0.083^\circ/\text{dan}$$

$$\frac{2\pi}{T} = 2\pi \sqrt{\frac{a^3}{Gm}} \Rightarrow n^2 a^3 = Gm$$

orbitalna brzina kretanja po elipsi:

$$v^2 = Gm \left(\frac{2}{r} - \frac{1}{a} \right)$$

$$E = \frac{v^2}{2} - \frac{Gm}{r}$$

Važi:

$$r = a(1 - e \cos E)$$

$$f = v$$

$$\tan\left(\frac{f}{2}\right) = \sqrt{\frac{1+e}{1-e}} \tan\left(\frac{E}{2}\right)$$

E - ekscentrična anomalija

$$M = n(t - t_0) = E - e \sin E$$

- Keplerova jednačina

srednja anomalija

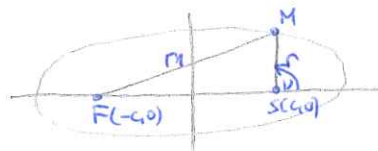
t_0 - posljednji prolaz kroz perihel

Keplerova jednačina nije riješiva analitički u opštem slučaju, ali se može analitički aproksimirati za malo e

$$f(E) = E - e \sin E - M(t)$$

$$E_{n+1} = E_n - \frac{f(E_n)}{f'(E_n)} = E_n - \frac{E_n - e \sin E_n - M(t)}{1 - e \cos E_n}$$

① Izvesti polarne jednadžbe elipse



$$r_1 + r_2 = 2a$$

$\triangle FSM$:

$$r_1^2 = r^2 + (FS)^2 - 2(FS)r \cos(180^\circ - u)$$

$$FS = 2c = 2ae$$

$$r_1^2 = r^2 + 4a^2e^2 - 4aer \cos u$$

$$4a^2 + r^2 = 4ar = r^2 + 4a^2e^2 - 4aer \cos u$$

$$a^2 - ar = a^2e^2 - aer \cos u$$

$$a^2(1 - e^2) = ar(1 + e \cos u)$$

b^2

$$r = \frac{b^2}{a}$$

$$\frac{1}{1 + e \cos u} \Rightarrow r = \frac{p}{1 + e \cos u}$$

p - parameter ellipse

$$\vec{F} = f \cdot \frac{m \cdot M}{r^2}$$

Ovaj zakon je opšten na dva tela:

$$\vec{F} = -f \cdot \frac{m \cdot M}{r^3} \cdot \vec{r}$$

$$F = f \cdot \frac{m \cdot M}{r^2}$$

Ovaj zakon se objašnjen kao kretanje planeta, poželja, razvoja, uticaja, spajanje, razdvajanje, putanja, osaka...

92) DIFERENCIJALNE JEONACIJE KRETANJA PLANETA

$\vec{g}_1 \rightarrow$ ubrzanje koje Sunce stvara na planetu

$$\vec{g}_1 = -f \cdot \frac{M}{r^3} \cdot \vec{r}$$

$\vec{g}_2 \rightarrow$ ubrzanje koje planeta stvara na sebi

$$\vec{g}_2 = f \cdot \frac{m}{r^3} \cdot \vec{r}$$

UBRZANJE PLANETE u odnosu na Sunce je $\vec{g} = \vec{g}_1 - \vec{g}_2$

$$\vec{g} = -f \cdot \frac{M+m}{r^3} \cdot \vec{r}, \quad \mu = f(M+m)$$

$$\vec{g} = -\mu \cdot \frac{\vec{r}}{r^3}$$

Postavimo vektor $\vec{r}$ u koordinatni sistem sa centrom u Suncevoj središtu:

$$g_x = -\mu \cdot \frac{x}{r^3}, \quad g_y = -\mu \cdot \frac{y}{r^3}, \quad g_z = -\mu \cdot \frac{z}{r^3}$$

Reka II. Njutnovom zakonu imamo $\vec{F} = m \cdot \vec{a}$

$$u = \frac{dr}{dt}, \quad a = \frac{dv}{dt} = \frac{dr}{dt} \Rightarrow \vec{F} = m \cdot \frac{d^2 \vec{r}}{dt^2}$$

$$F_x = m \cdot \frac{d^2 x}{dt^2}, \quad F_y = m \cdot \frac{d^2 y}{dt^2}, \quad F_z = m \cdot \frac{d^2 z}{dt^2}$$

Osnovni zadatak mehanike je da se odredi kretanje tjele u polju gravitacije. To su diferencijalne jednačine.

$$\begin{aligned} \frac{d^2 x}{dt^2} &= -\mu \cdot \frac{x}{r^3} \\ \frac{d^2 y}{dt^2} &= -\mu \cdot \frac{y}{r^3} \\ \frac{d^2 z}{dt^2} &= -\mu \cdot \frac{z}{r^3} \end{aligned}$$

93) IZVOĐENJE II. KEPLEROVOSI ZAKONA

Da bismo dobili kretanje planete oko Sunca po zakonu gravitacije trebamo složiti dif. jednačije:

$$\begin{aligned} \frac{d^2 x}{dt^2} &= -\mu \cdot \frac{x}{r^3} \quad | \cdot (x^2) \cdot \Delta \\ \frac{d^2 y}{dt^2} &= -\mu \cdot \frac{y}{r^3} \quad | \cdot (y^2) \cdot \Delta \\ \frac{d^2 z}{dt^2} &= -\mu \cdot \frac{z}{r^3} \quad | \cdot (z^2) \cdot \Delta \end{aligned}$$

$$\begin{aligned} x \cdot \frac{d^2 x}{dt^2} - y \cdot \frac{d^2 y}{dt^2} &= 0 \\ y \cdot \frac{d^2 y}{dt^2} - z \cdot \frac{d^2 z}{dt^2} &= 0 \\ z \cdot \frac{d^2 z}{dt^2} - x \cdot \frac{d^2 x}{dt^2} &= 0 \end{aligned}$$

Ovakvo dobijeni izrazi su zasnovani na integriranju:

$$\begin{aligned} x \cdot \frac{d^2 x}{dt^2} - y \cdot \frac{d^2 y}{dt^2} &= C \quad | \cdot (-1) \Rightarrow \frac{d}{dt} \left(x \cdot \frac{dx}{dt} - y \cdot \frac{dy}{dt} \right) = \frac{dC}{dt} = 0 \\ y \cdot \frac{d^2 y}{dt^2} - z \cdot \frac{d^2 z}{dt^2} &= A \quad | \cdot (-1) \Rightarrow \frac{d}{dt} \left(y \cdot \frac{dy}{dt} - z \cdot \frac{dz}{dt} \right) = \frac{dA}{dt} = 0 \\ z \cdot \frac{d^2 z}{dt^2} - x \cdot \frac{d^2 x}{dt^2} &= B \quad | \cdot (-1) \Rightarrow \frac{d}{dt} \left(z \cdot \frac{dz}{dt} - x \cdot \frac{dx}{dt} \right) = \frac{dB}{dt} = 0 \end{aligned}$$

$$z \cdot x \frac{dx}{dt} - 2y \frac{dy}{dt} = C_2$$

$$\frac{dx}{dt} - x \cdot 2 \cdot \frac{dy}{dt} = Ax \Rightarrow \frac{Ax + By + Cz = 0}{\text{JEDNAČINA RAVNI - PUTANJA}}$$

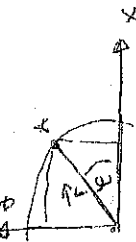
PLANETE JE KRIVA ČIJA RAVAN PROLAZI KROZ SREDIŠTE SUNCIA

AKO ZA TU RAVAN UZMEHO X I Y RAVAN, KRETANJE PLANETE U NJOG JE ODREĐENO SA DVE JEDNAČICE:

$$\frac{dx}{dt} = -1 \cdot \frac{x}{r^3} \quad \frac{dy}{dt} = -1 \cdot \frac{y}{r^3}$$

$$x \frac{dx}{dt} - y \frac{dy}{dt} = 0$$

$$\int (x \frac{dx}{dt} - y \frac{dy}{dt}) = \int x \frac{dx}{dt} - y \frac{dy}{dt} = C_1$$



$$x = r \cos \varphi \quad y = r \sin \varphi$$

$$r \cos \varphi \cdot \frac{d}{dt}(r \sin \varphi) - r \sin \varphi \cdot \frac{d}{dt}(r \cos \varphi) = C_1$$

$$r \cos \varphi \cdot (r \sin \varphi + r \cos \varphi \frac{d\varphi}{dt}) - r \sin \varphi \cdot (r \frac{d\varphi}{dt} \cos \varphi - r \sin \varphi \frac{d\varphi}{dt}) = C_1$$

$$\sqrt{r^2 \frac{d\varphi}{dt} = C_1}$$

DVOSTRANA SEKTORSKA BRZINA (I KOPERNIKOVA)
DA BI U PREDSTAVLJAO 2, X - OSU TREBA ORJENTISATI
TAKO DA PROLAZI KROZ PERIHEL.

34. IZVOĐENJE I KOPERNIKOVSA ZAKONA

DA BISMO GLELI I KOPERNIKOV ZAKON TREBA DA NADJEHO TRAJEKTORE U PLANETI:

$$\frac{dx}{dt} = -1 \cdot \frac{x}{r^3} \quad \frac{dy}{dt} = -1 \cdot \frac{y}{r^3}$$

$$\frac{dx}{dt} \cdot \frac{dx}{dt} + \frac{dy}{dt} \cdot \frac{dy}{dt} = -\frac{1}{r^3} (x \cdot \frac{dx}{dt} + y \cdot \frac{dy}{dt})$$

$$\vec{r} = x \cdot \vec{i} + y \cdot \vec{j} \quad r \cdot \frac{d\vec{r}}{dt} = x \cdot \frac{d\vec{i}}{dt} + y \cdot \frac{d\vec{j}}{dt} \quad r^2 = x^2 + y^2$$

$$\frac{d}{dt}(r^2) = \frac{d}{dt}(x^2 + y^2)$$

$$2r \frac{dr}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$r^2 = x^2 + y^2 = (\frac{dx}{dt})^2 + (\frac{dy}{dt})^2$$

$$\frac{1}{2} \frac{d}{dt} ((\frac{dx}{dt})^2 + (\frac{dy}{dt})^2) = -\frac{1}{r^3} (x \cdot \frac{dx}{dt} + y \cdot \frac{dy}{dt})$$

$$\frac{1}{2} \frac{d}{dt}(r^2) = -\frac{1}{r^3} r \cdot \frac{dr}{dt} \quad \frac{d}{dt}(\frac{1}{r}) = -\frac{1}{r^2} (\frac{dr}{dt})$$

$$\frac{d}{dt}(r^2) = 2r \cdot \frac{dr}{dt} (\frac{1}{r})$$

INTEGRACIJA => $\int \frac{1}{r^2} \frac{dr}{dt} dt = C_2$ OVAJ INTEGRAL SE LAKE INTEGRALIZIRAJE

VAJEDNOST KONSTANTE C2 ODREĐUJE SE IZ POZNATIH USLOVA KRETANJA (RADIJUS-VEKTORA I BRZINE U TREKUTKU t0)

$$x = r \cos \varphi$$

$$y = r \sin \varphi$$

$$v^2 = (\frac{dx}{dt})^2 + (\frac{dy}{dt})^2 = (\frac{d}{dt}(r \cos \varphi))^2 + (\frac{d}{dt}(r \sin \varphi))^2$$

$$v^2 = (\frac{dr}{dt} \cos \varphi - r \frac{d\varphi}{dt} \sin \varphi)^2 + (\frac{dr}{dt} \sin \varphi + r \frac{d\varphi}{dt} \cos \varphi)^2$$

$$v^2 = (\frac{dr}{dt})^2 \cos^2 \varphi - 2r \frac{dr}{dt} \frac{d\varphi}{dt} \sin \varphi \cos \varphi + (\frac{dr}{dt})^2 \sin^2 \varphi + (\frac{dr}{dt})^2 \sin^2 \varphi + 2r \frac{dr}{dt} \frac{d\varphi}{dt} \sin \varphi \cos \varphi + (\frac{dr}{dt})^2 \cos^2 \varphi$$

35. Изведем III Кеплеров закон

Планета воит се вращает по кругу с радиусом r и периодом T .
 Дано: r , T .
 Найти: M .

$$Q = \frac{2\pi r}{T}$$

$$v = \frac{2\pi r}{T} \Rightarrow v^2 = \frac{4\pi^2 r}{T^2}$$

$$\frac{v^2}{r} = \frac{4\pi^2}{T^2}$$

$$\frac{v^2}{r} = \mu \Rightarrow \frac{4\pi^2}{T^2} = \mu$$

$$\mu = f(M, m)$$

$$M \rightarrow \text{масса Солнца}$$

$$m \rightarrow \text{масса планеты}$$

$$\frac{4\pi^2}{T^2} = \mu \Rightarrow \frac{4\pi^2}{T^2} = \frac{GM}{r^2} \Rightarrow \frac{4\pi^2}{T^2} = \frac{GM}{r^2} \Rightarrow \frac{4\pi^2}{T^2} = \frac{GM}{r^2}$$

$$\frac{4\pi^2}{T^2} = \mu \Rightarrow \frac{4\pi^2}{T^2} = \frac{GM}{r^2} \Rightarrow \frac{4\pi^2}{T^2} = \frac{GM}{r^2}$$

$$\frac{I_1(M+m)}{I_2(M+m)} = \frac{a_1^3}{a_2^3}$$

Помогает третий Кеплеров закон можно определить массу планеты.

$$\frac{I_1^2 (M+m)}{I_2^2 (M+m)} = \frac{a_1^3}{a_2^3}$$

$$M = \left(\frac{a_1^3}{a_2^3} \right) \cdot \left(\frac{I_2}{I_1} \right)^2 \cdot M_0$$